

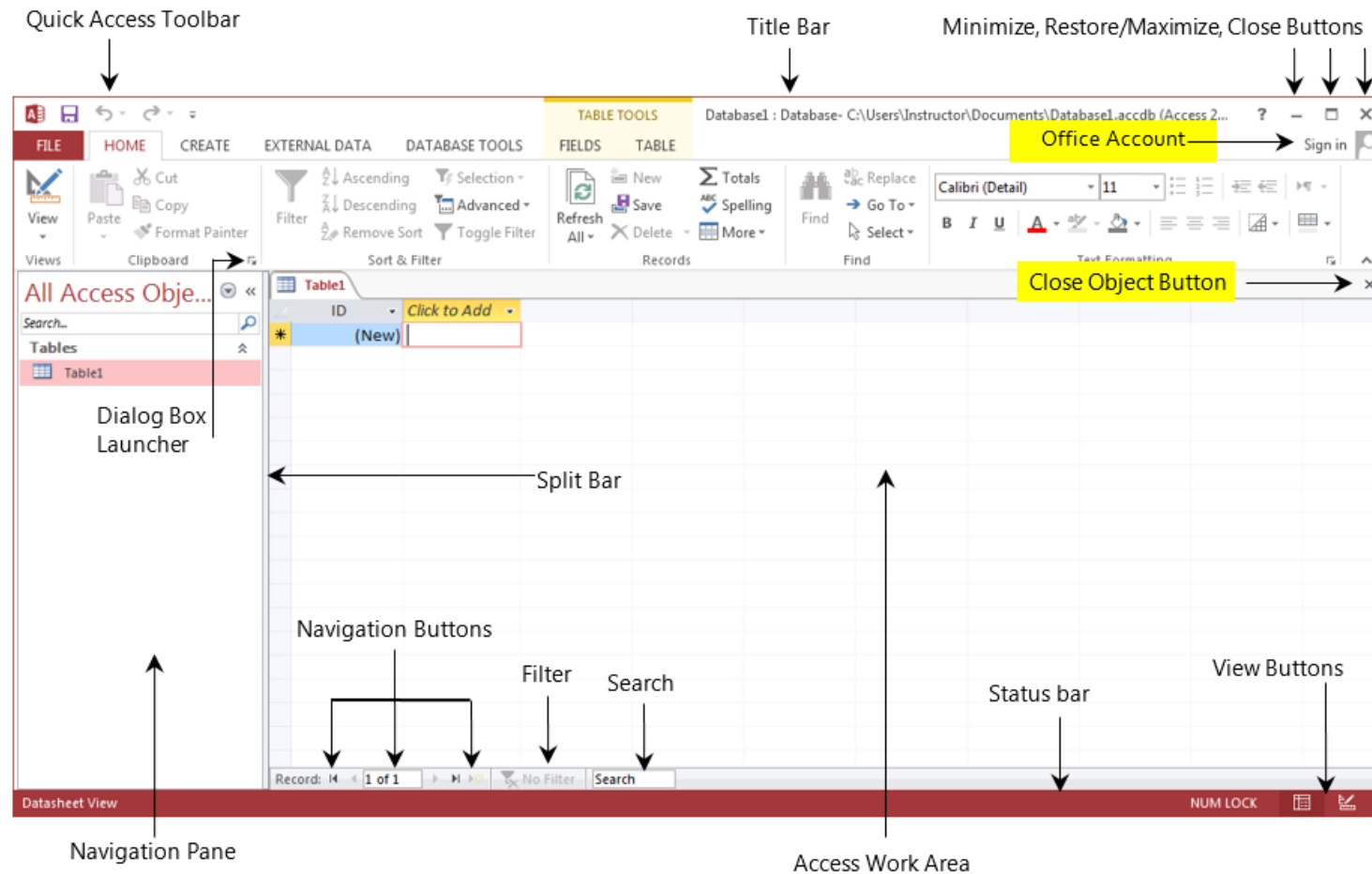
Advanced MSOffice/OpenOffice (Word, Excel, Access, PPT)

Microsoft Access

What is Microsoft Access?

- Relational Database Management System (RDBMS),
- designed primarily for home or small business usage.
- desktop database system because its functions are intended to be run from a single computer.
- in contrast to a server database application (such as SQL Server), where it is intended to be installed on a server, then accessed remotely from multiple client machines.
 - Includes the tools you need to develop user-friendly interfaces and reports
 - Provides a user-friendly environment for accomplishing database tasks even without prior knowledge
 - Uses SQL code behind the scenes

Microsoft Access opening Screen

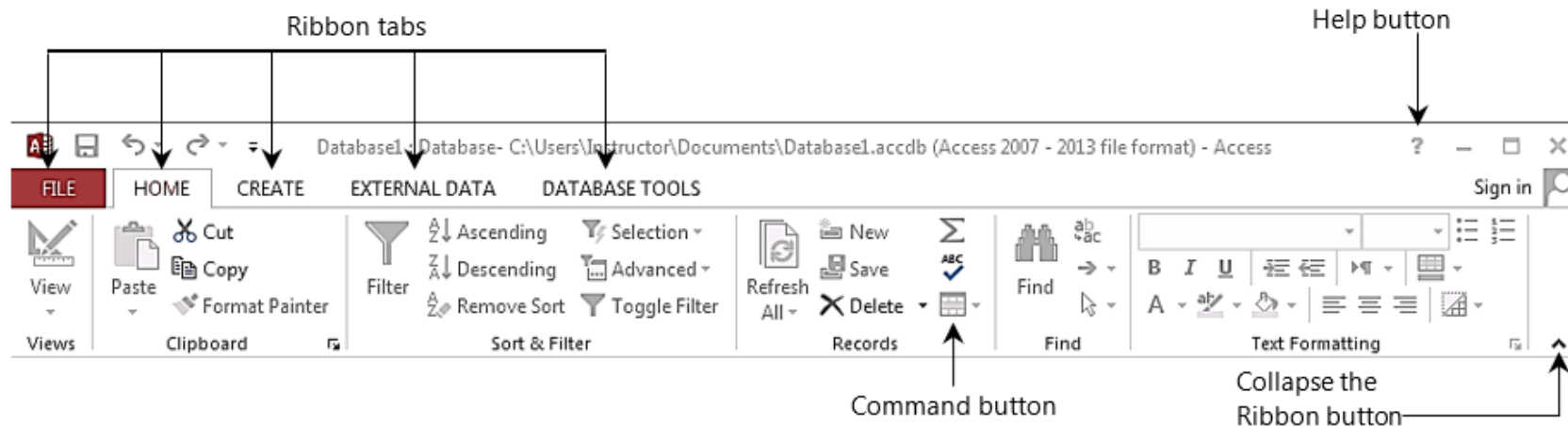


Components of the Microsoft Access User Interface

- Ribbon
- Navigation Pane
- Backstage view

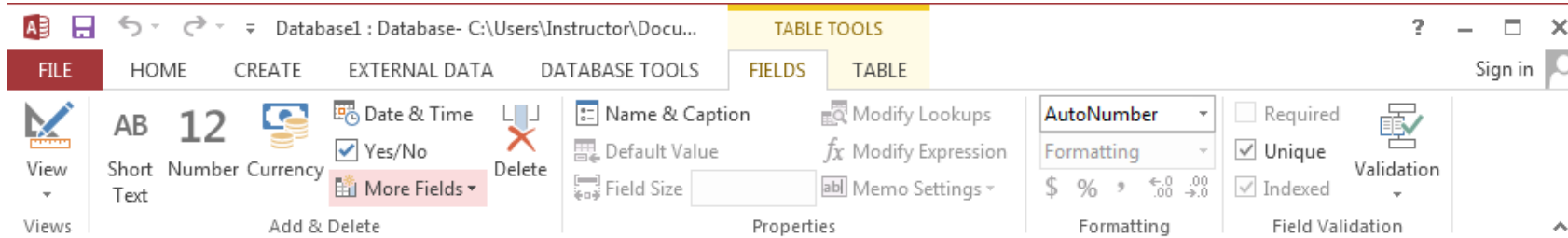
The Ribbon

The Ribbon contains the commands and options you use to perform database tasks. Each tab contains a group of related commands.



The Ribbon(Cont.)

To reduce screen clutter, contextual tabs appear only when they are applicable. For example, the Table Tools tab appears only when a table is open.

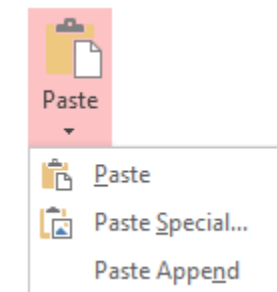
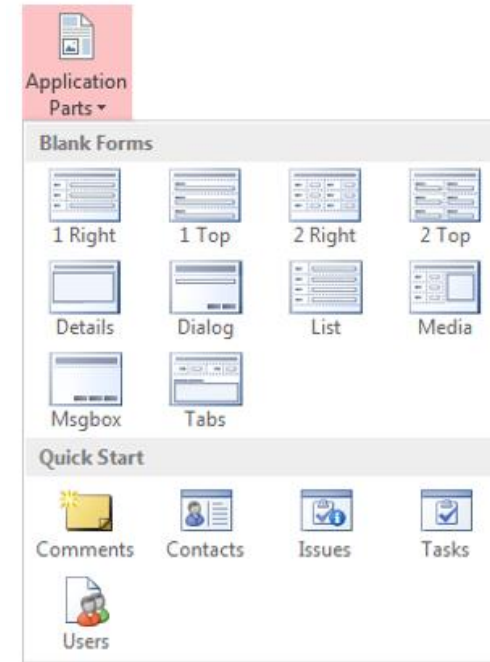


The Ribbon includes:

Ribbon tabs	Relate to a specific type of activity. You will see contextual tabs when they are applicable.
Command buttons	Appear on the ribbon tabs in groups. When you click a command button, an action takes place.
Collapse the Ribbon button (CTRL+F1)	Collapse the Ribbon to just the tab names in order to conserve space on the screen. You can still use the Ribbon while it is collapsed. To toggle between a collapsed and expanded state, press Ctrl+F1.
Microsoft Access Help (F1) button	Provides access to the Help system.

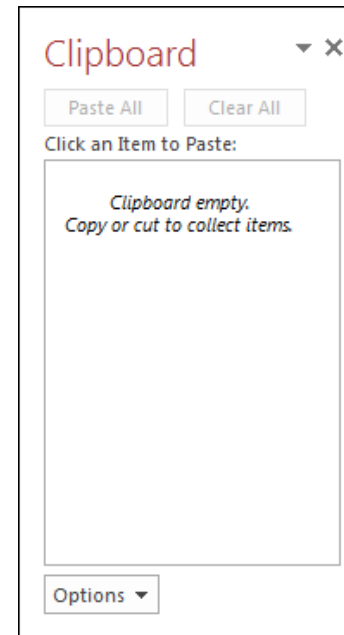
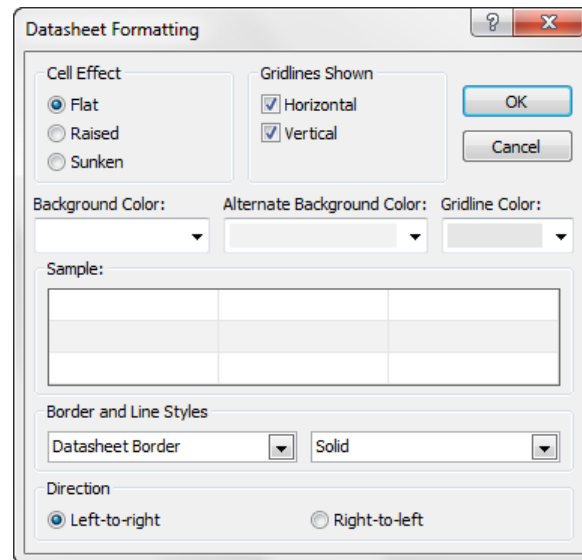
The Ribbon

Each tab on the Ribbon contains groups with similar commands. If a group shows a feature with a drop-down arrow, click the arrow to display a list or a gallery of options for that feature.

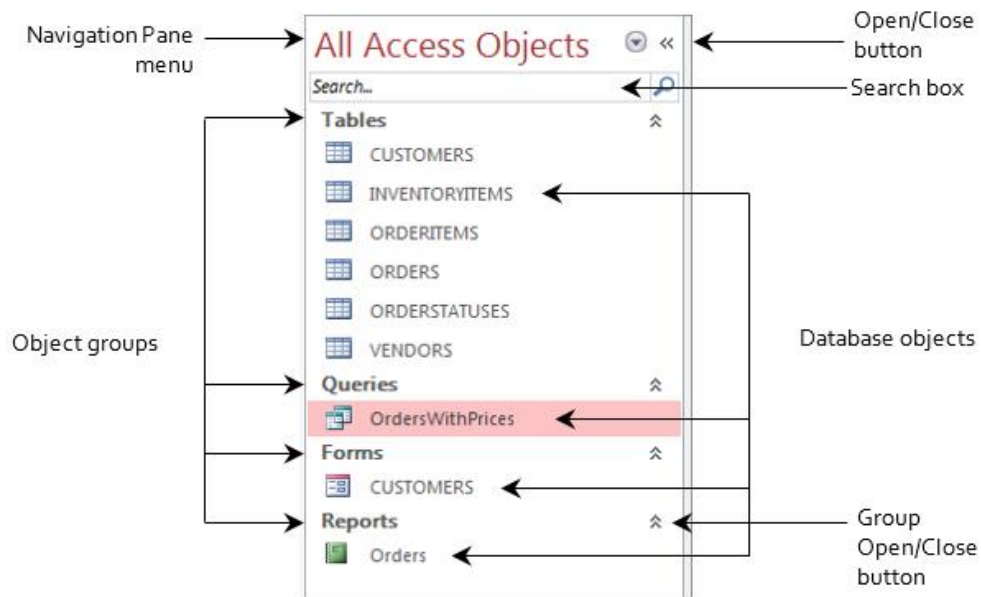


The Ribbon

Click the dialog box launcher at the lower right of a group to display a dialog box, window or task pane



The Navigation Pane

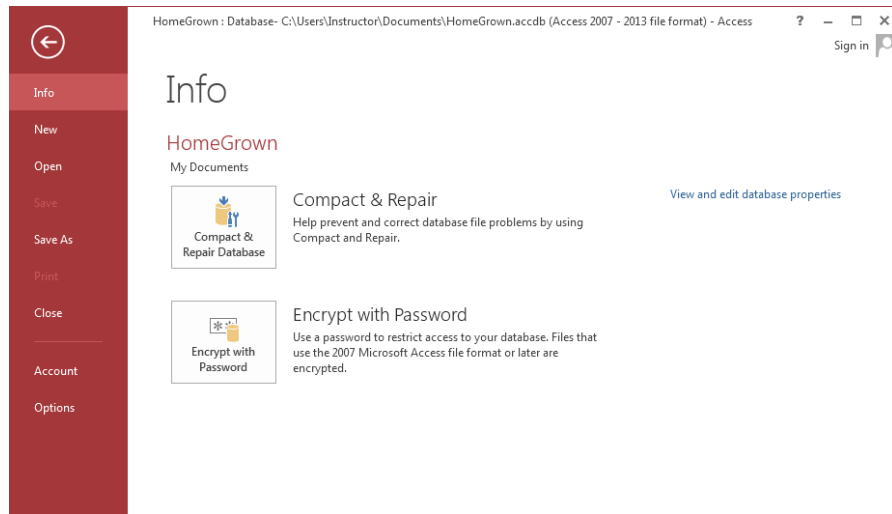


- The Navigation Pane is a central location from which you can use any type of database object
- The Navigation Pane replaces the Database window found in versions of Access previous to 2007.

The Navigation Pane includes:

Navigation Pane menu	Click the drop-down arrow to open a menu that allows you to specify which objects to show. The default selection is All Access Objects.
Open/Close buttons	Collapses the Navigation Pane so only its Open/Close button and shutter bar display.
Object groups	Organize the objects in a database so you can find what you are looking for.
Database objects	The tables, queries, forms and other objects that exist in the database. Each object appears in its appropriate group.
Search box	You can type the name of an object you want to find

Backstage View



- Used to create, open, save, print and manage databases
- Includes commands and options related to things you can do with a database itself
- Shows the commands and options available on the File tab of the Ribbon

The screen includes the following components:

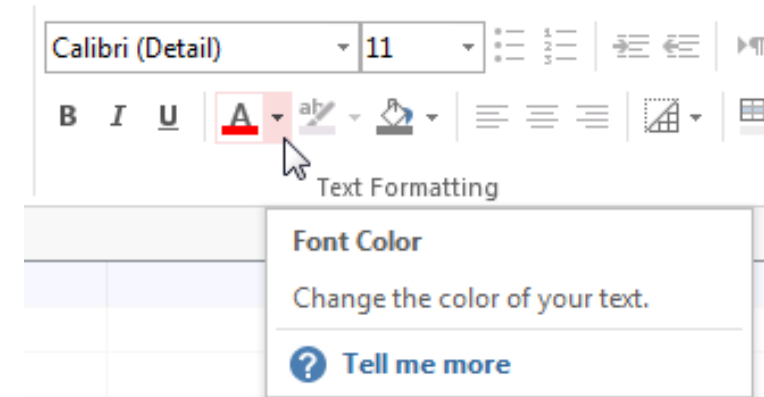
Title bar	Lists the name of the open file and its path
Quick Access Toolbar	Provides access to frequently used commands. By default, includes the Save , Undo and Redo commands.
Minimize, Restore/Maximize and Close buttons	Enable you to minimize the program to a button on the taskbar, maximize the window to full screen, restore the window to its former size, and close the application
Close object button	Closes the object viewed on the right of the Navigation Pane
Dialog box launcher	Opens a dialog box, window or task pane when you click it
Access work area	When objects in the Navigation Pane are opened, they appear in the Access work area so you can work with them
View buttons	Enable you to switch between views of the currently selected object

The screen includes the following components:

Status bar	Indicates the currently displayed view and whether specific modes are applied.
Split bar	Drag this bar to increase or decrease the size of the Navigation Pane, or to increase or decrease the size of the object you are viewing in the work area.
Navigation buttons	Enable you to move between records in a table or form.
Filter	Enables you to remove or re-apply a filter
Search	Enables you to search and find characters in the table records.
Office Account	Sign in to Office (optional) to save your files online to a SkyDrive account.

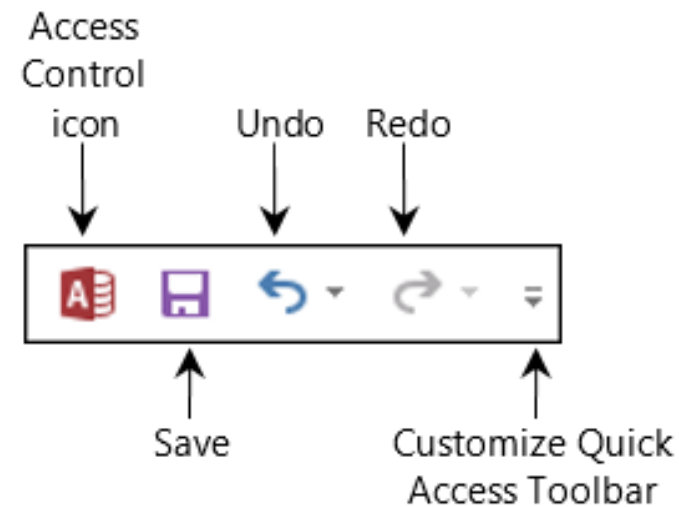
Viewing Screen Tips

- ScreenTips help identify buttons or elements on the tabs of the Ribbon and the screen
- To view a ScreenTip, position the mouse pointer over the item; a tip then displays the name of the button along with a description of its purpose.



Using the Quick Access Toolbar

- Includes buttons for frequently used commands
- By default, includes the Save, Undo, Redo and Customize Quick Access Toolbar buttons
- By default, located at the left of the title bar above the Ribbon
- You can move the Quick Access toolbar and you can customize it by adding buttons



Access Database File

- An Access database file is a container for a variety of database objects
- All database objects are stored within a single database file with the .accdb file name extension
 - Access 2007 also uses the .accdb extension
 - Earlier versions of Access use the .mdb extension

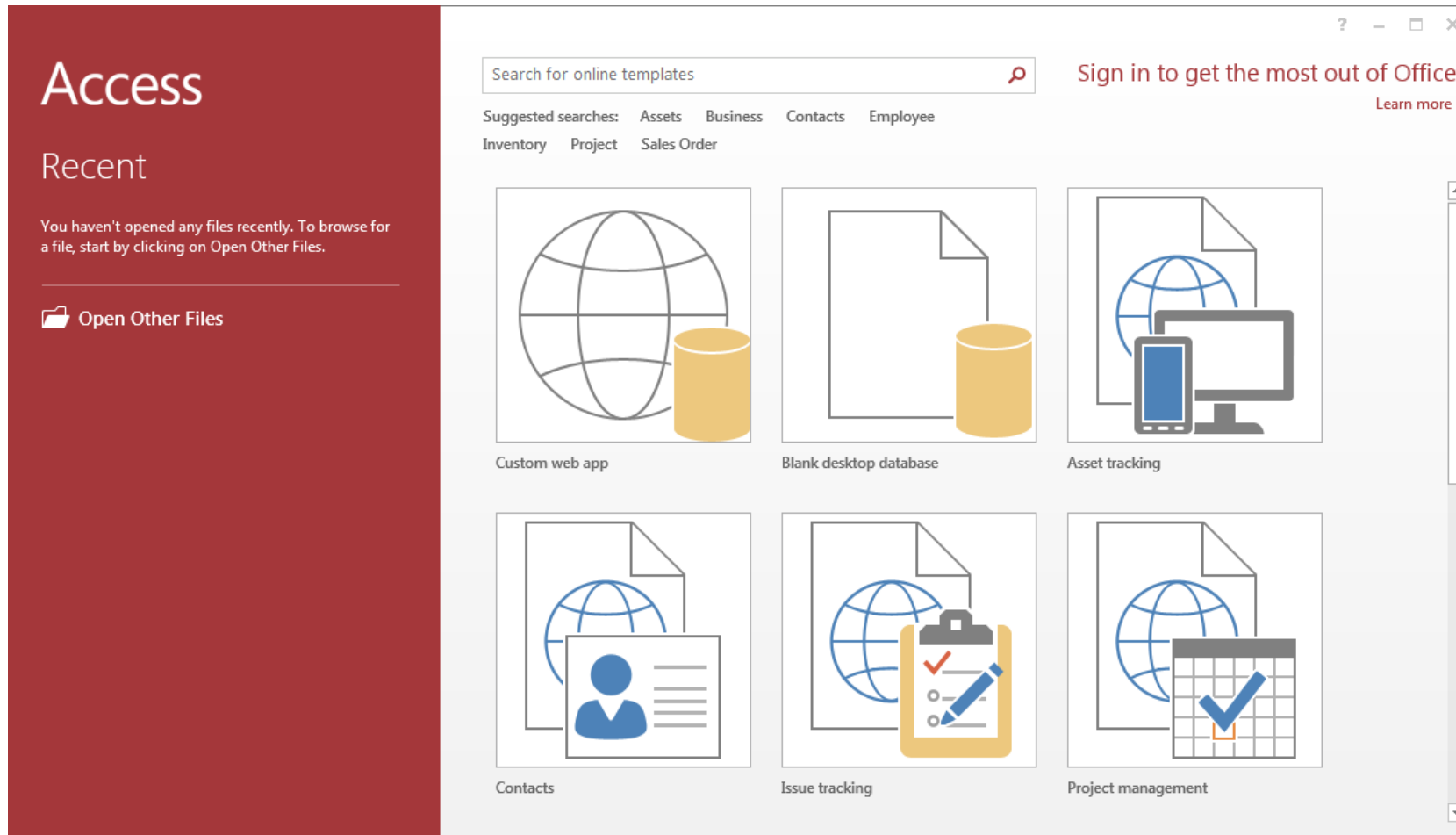
Database Objects

- Tables – store data in rows and columns. In a table, each row is a record, and each record consists of one or more fields
- Forms – function as user-friendly interfaces for working with data
- Reports – print and summarize data
- Queries – extract subsets of data from one or more tables, allowing you to view exactly the data you want
- Macros – add functionality to database objects and are often used to automate certain database tasks
- Modules – also add functionality to a database, and are stored collections of Visual Basic for Applications (VBA) programming language statements and procedures

Desktop Databases and Access Apps

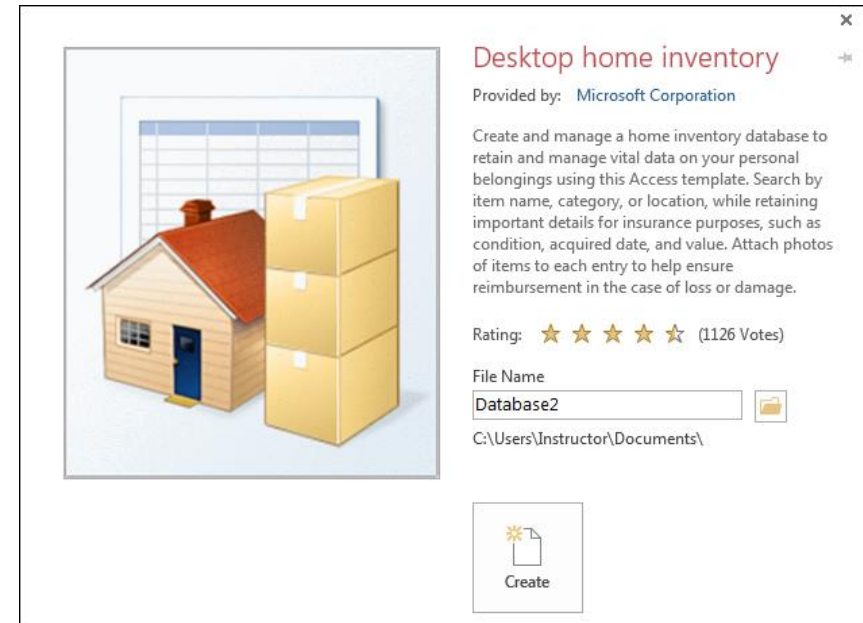
- An Access app is a new type of database that you create, design and modify in Access and then use in a standard web browser
- You share Access apps with other users as SharePoint apps
- To build an app, select the type of data you want to track and Access creates the database structure for you

Starting Access



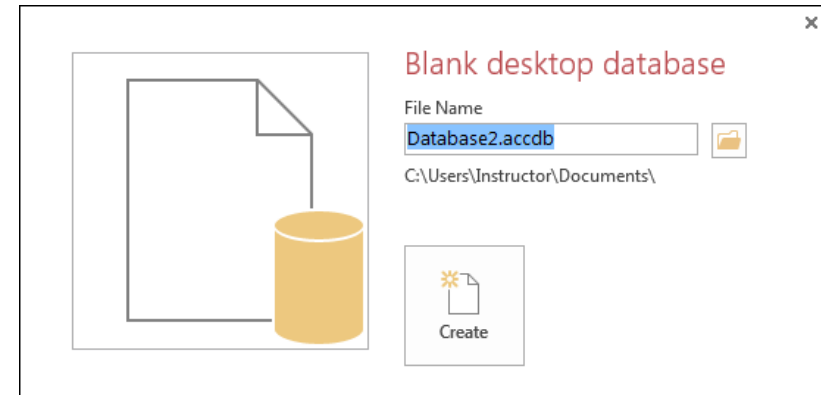
Wizards and Templates

- Database templates have replaced the Database Wizard found in older versions
- Some templates include objects and sample data.
- The Blank database template does not include any pre-defined database objects



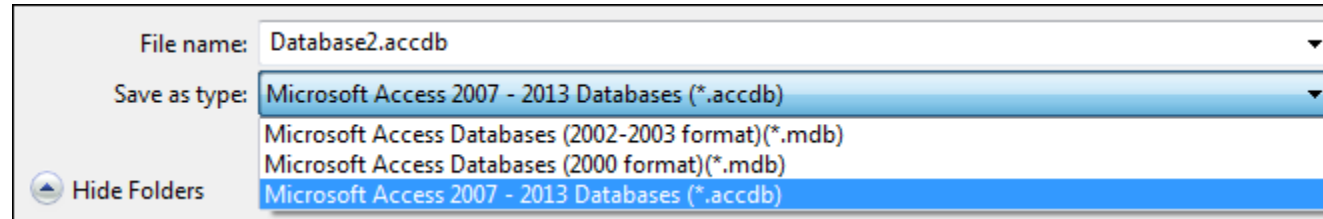
Creating a New Blank Database

- The Blank database template creates an empty database file
- Click the **Browse for a location** button to specify a name and location for the new database
- Specify a file name and location and click **OK** to save your specifications
- Click the **Create** button to create the blank database file



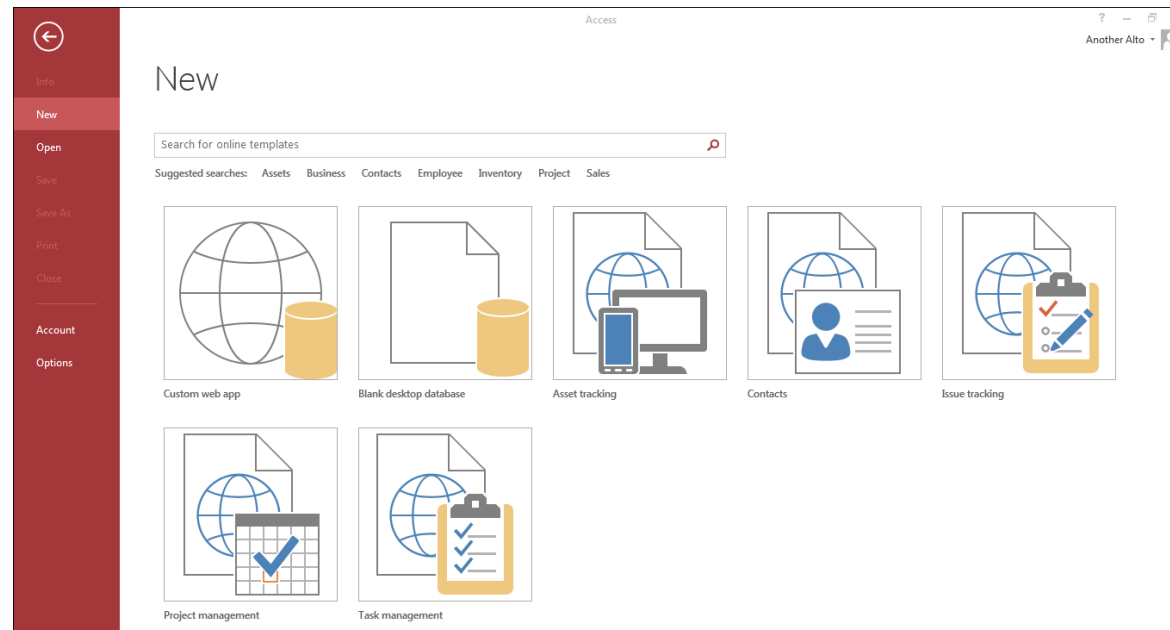
Creating Databases in Older Formats

- To save a database in an older file type, display the Save as type drop-down list in the File New Database dialog box



Creating a Database from a Template

- Click a template in the gallery
- Specify a name and location
- Click **Create**

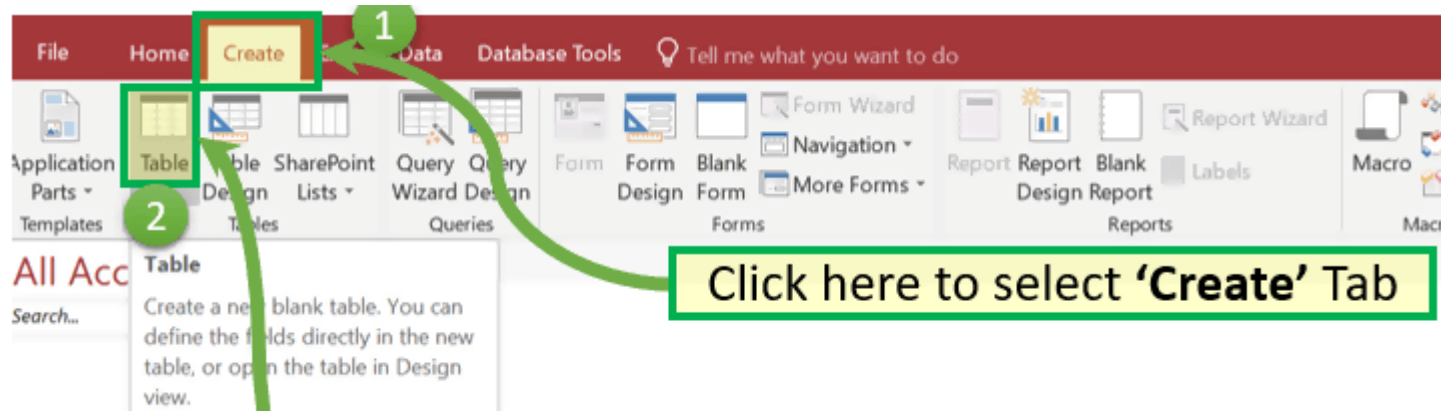


Creating a Table

- Tables store data
- Post creation of the table, we can keep inserting the rows in the table.
- There are two ways to create Table in MS Access
 - Create a Table from **Design View**
 - Create a Table from **Datasheet View**

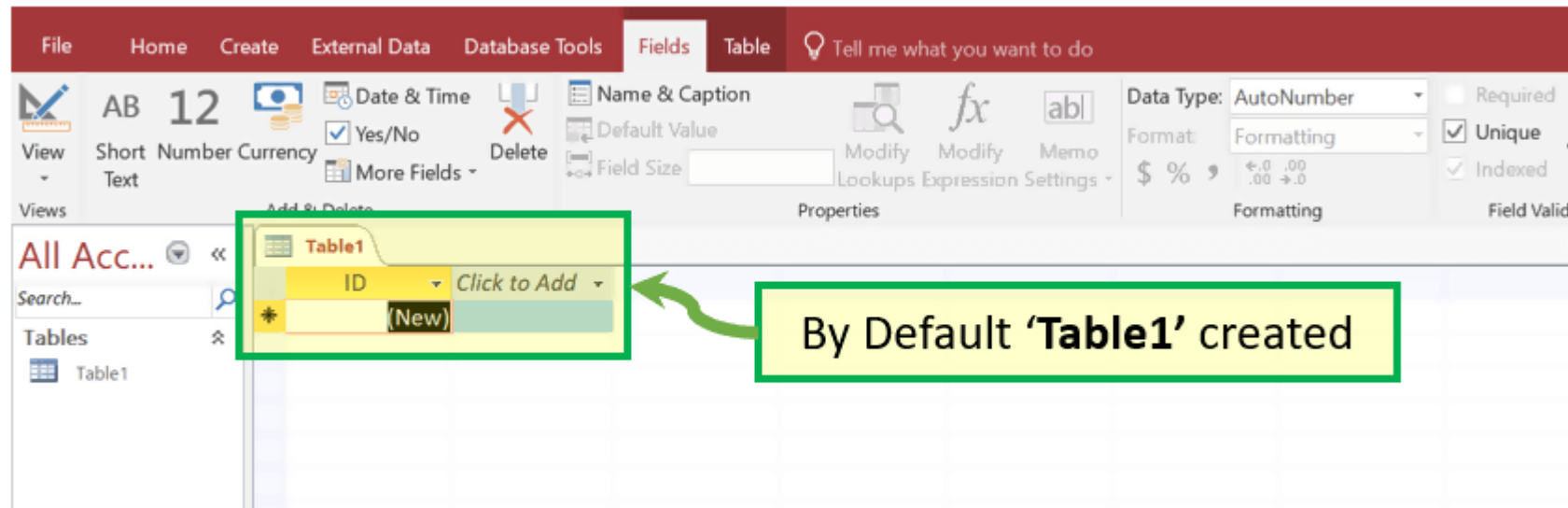
Create Table – Datasheet View

- **Step 1)** First Click **Create** tab. Then from **Tables** group, click **Table**.



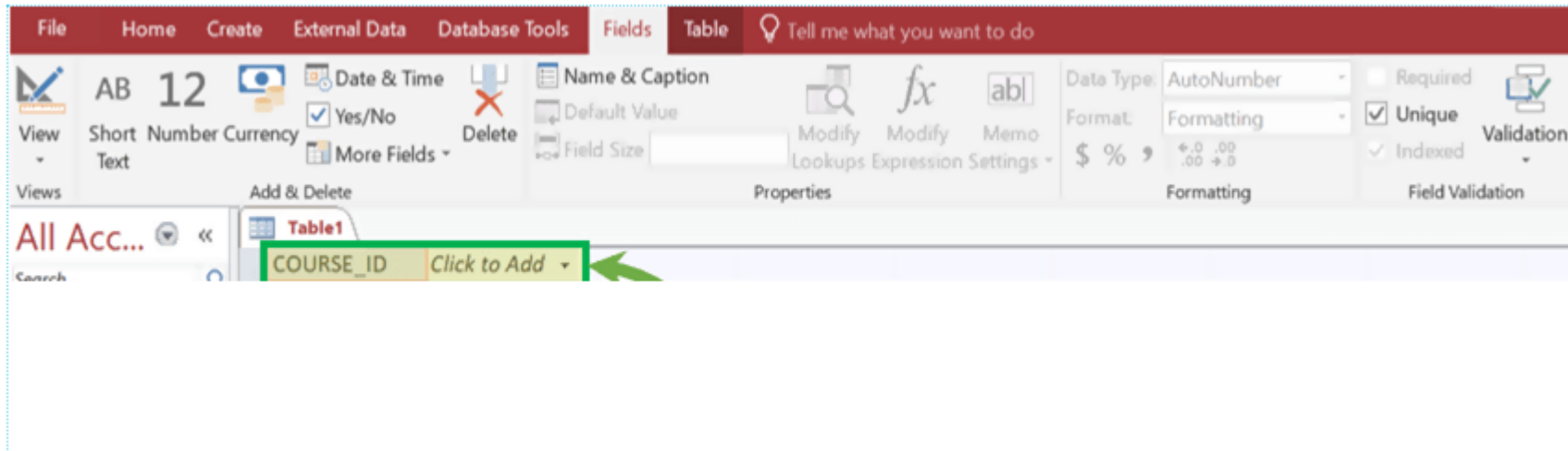
Create Table – Datasheet View

- **Step 2)** System will display the default table created with 'Table1' name.



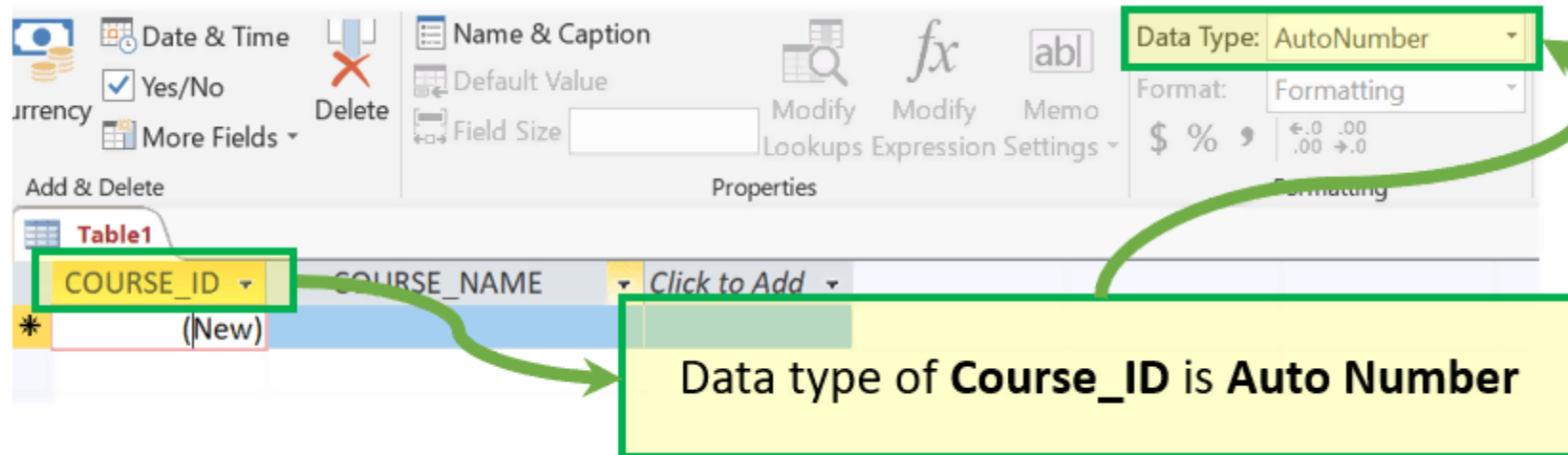
Create Table – Datasheet View

- **Step 3) To Rename Column**, double click on Column Header and enter the new column Name.



Create Table – Datasheet View

- Note that the Data type of Course_ID is '**AutoNumber**.' Hence this is also the Unique Key of the table.



Understanding Fields and Their Data Types

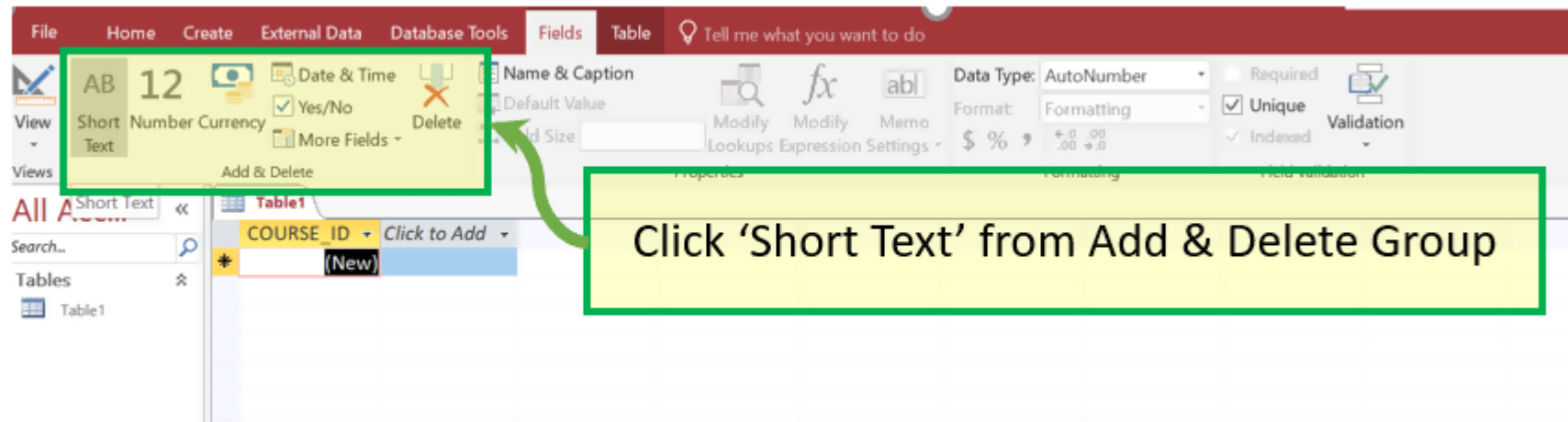
- Field - an element of a table that contains a specific item of information, such as a last name.
- Field's Data Type - determines what kind of data the field can store.

Understanding Fields and Their Data Types

Format	Use to display
Short Text	Alphanumeric data (names, titles, etc.) - up to 255 characters
Long Text	Large amounts of alphanumeric data: sentences and paragraphs – 64,000 characters
Number	Numeric data
Date/Time	Dates and times
Currency	Monetary values
AutoNumber	Unique value generated by Access for each new record
Yes/No	Yes and No values and fields that contain only one of two values
OLE Object	Pictures, graphs, or other ActiveX objects from another Windows-based application
Hyperlink	A link address to a document or file on the Internet
Attachment	You can attach files such as pictures, documents, spreadsheets, or charts; each Attachment field can contain an unlimited number of attachments per record, up to the storage limit of the size of a database file.
Calculated	You can create an expression that uses data from one or more fields. You can designate different result data types from the expression.
Lookup Wizard	Displays either a list of values that is retrieved from a table or query, or a set of values that you specified when you created the field. The Lookup Wizard starts and you can create a Lookup field. The data type of a Lookup field is either text or number, depending on the choices that you make in the wizard.

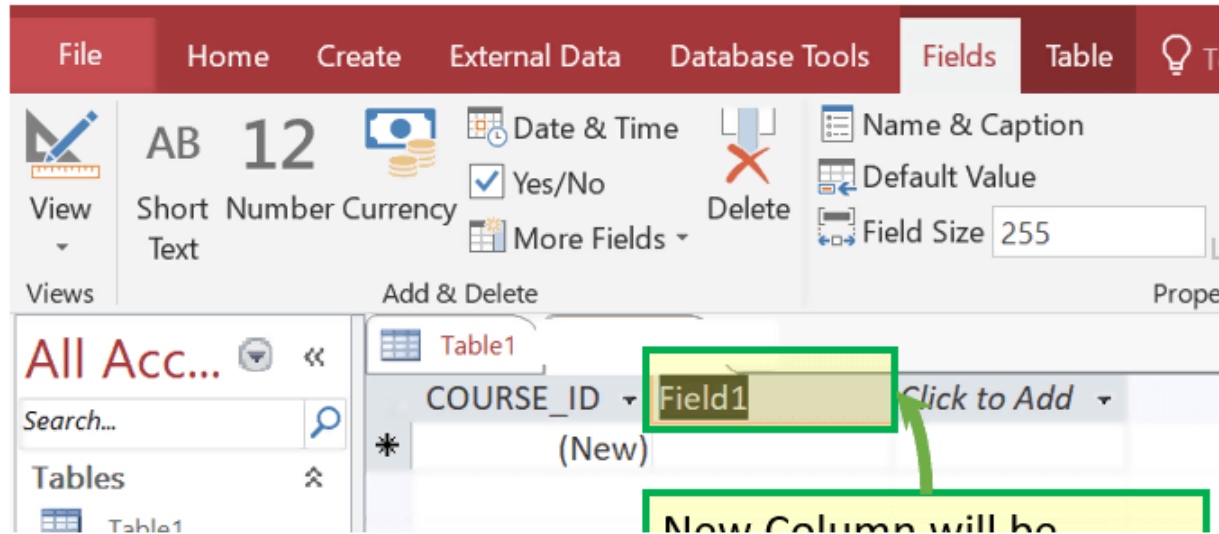
Create Table – Datasheet View

- **Step 4)** You can **Add Column** by clicking on any category from **the 'Add & Delete'** group. Alternatively, you can also add a column by clicking on 'Click to Add'
- For Example, click on 'Short Text' from the 'Add & Delete' group.



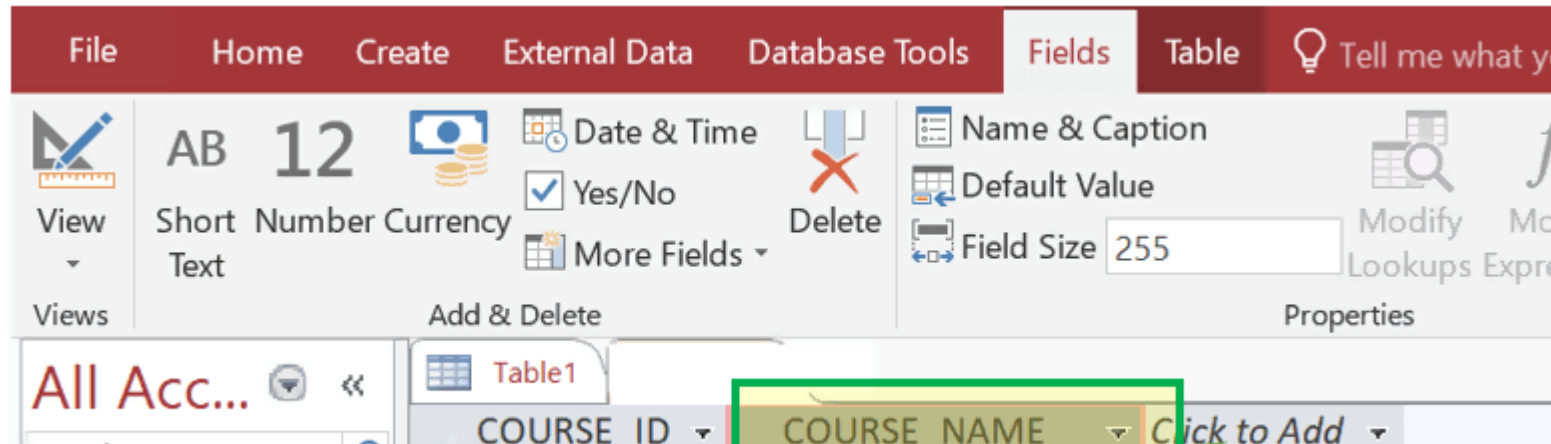
Create Table – Datasheet View

- **Step 5)** Column will be added with the default name as 'Field1.'



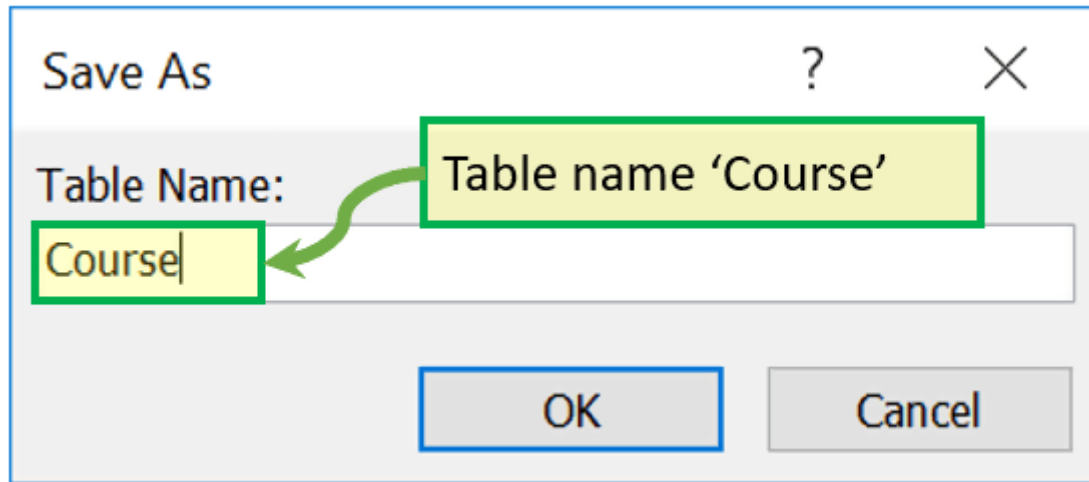
Create Table – Datasheet View

- **Step 6)** Click on Header and rename as **'COURSE_NAME.'**



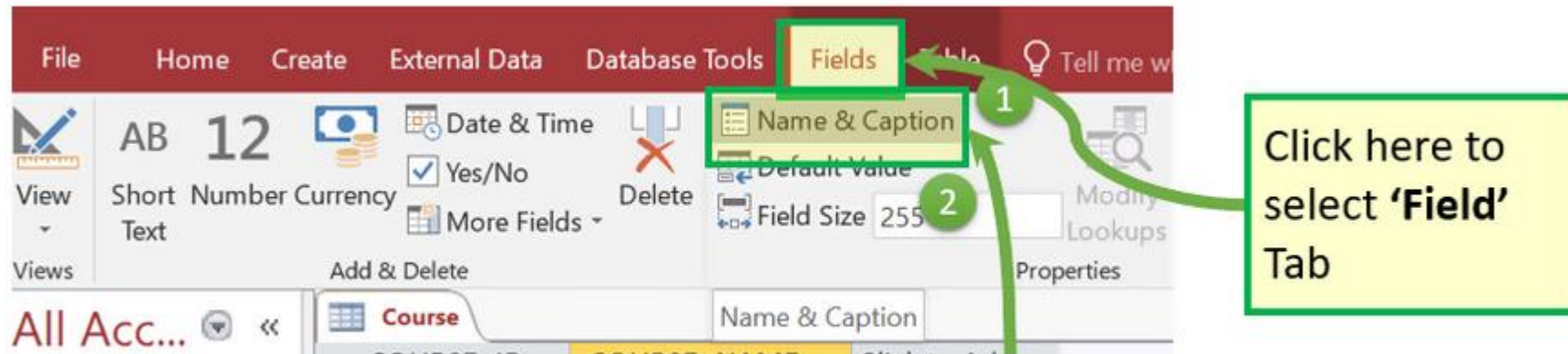
Create Table – Datasheet View

- **Step 7)** Press 'Ctrl + S' and Enter the new table name to save the table.



Create Table – Datasheet View

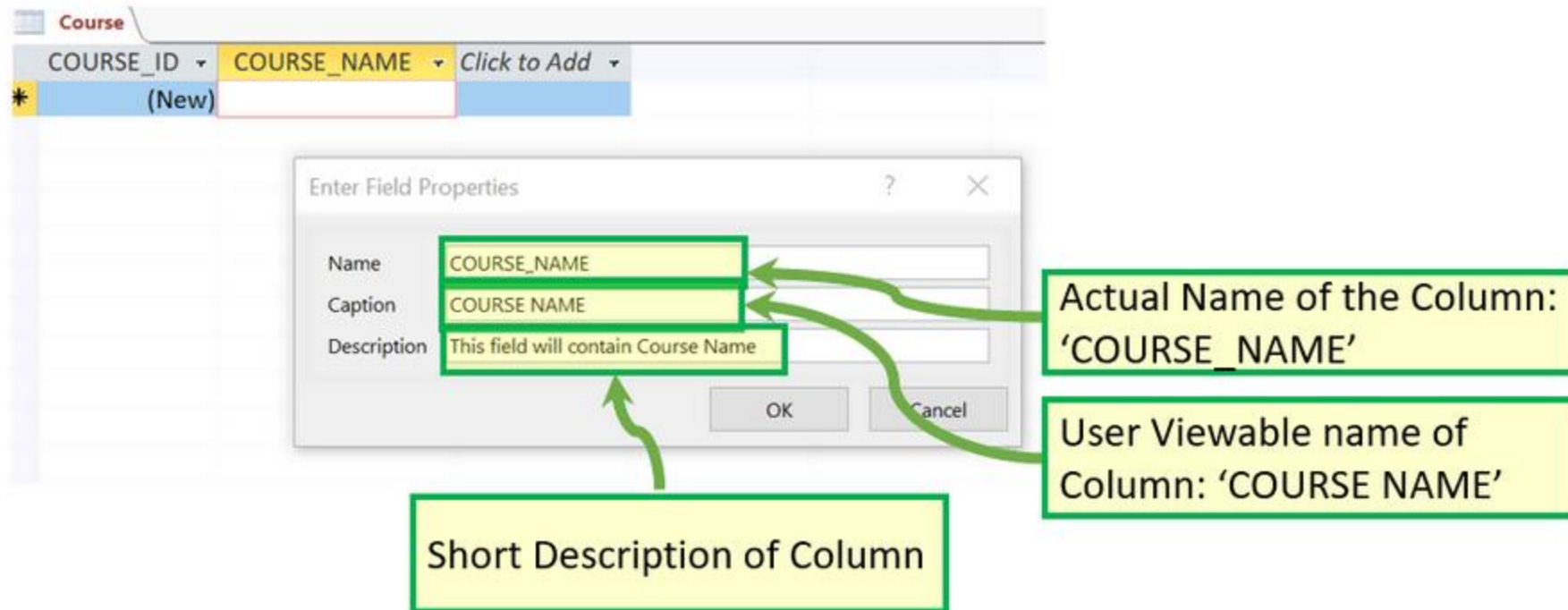
- **Step 8)** You can also save new Name, Caption and Short description for any Column by clicking on '**Name and Caption.**' Click on it



Create Table – Datasheet View

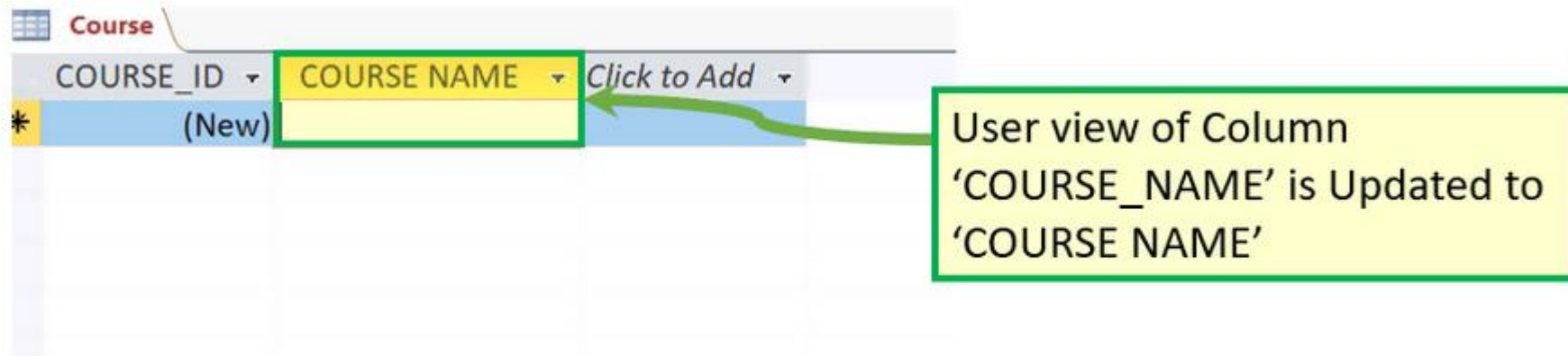
- **Step 9)** Dialog Box will appear. Add the following and Click on 'OK'
- 'Name' - This is the actual name of the column.
- 'Caption' - This is the user view name of the column.
- 'Description' - This is the short description of the column name

Create Table – Datasheet View



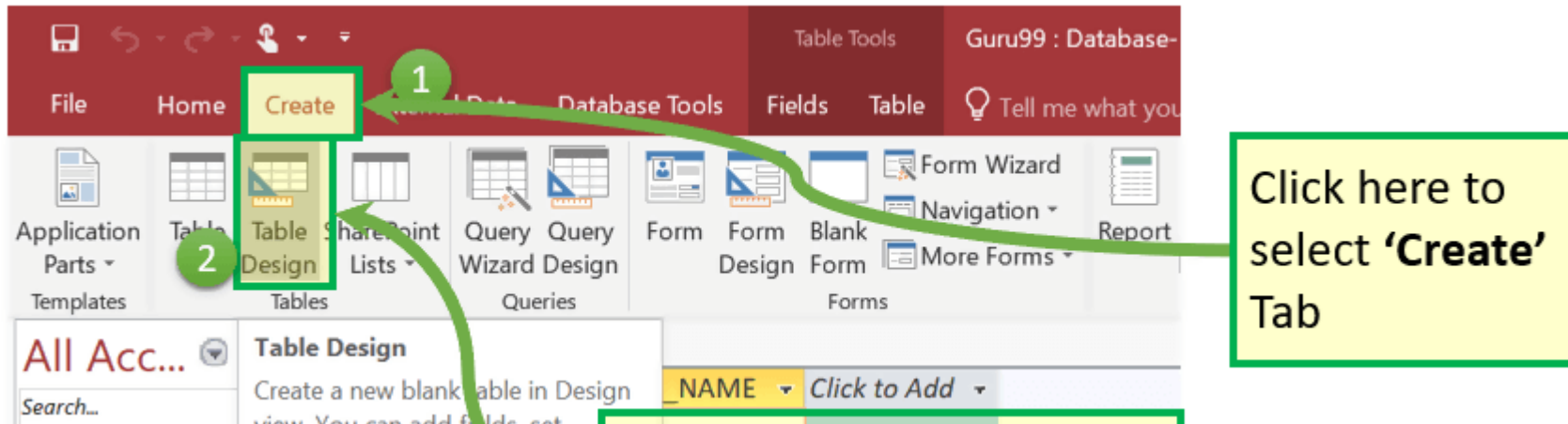
Create Table – Datasheet View

- **Result:** Name, Caption, and Description now exist.



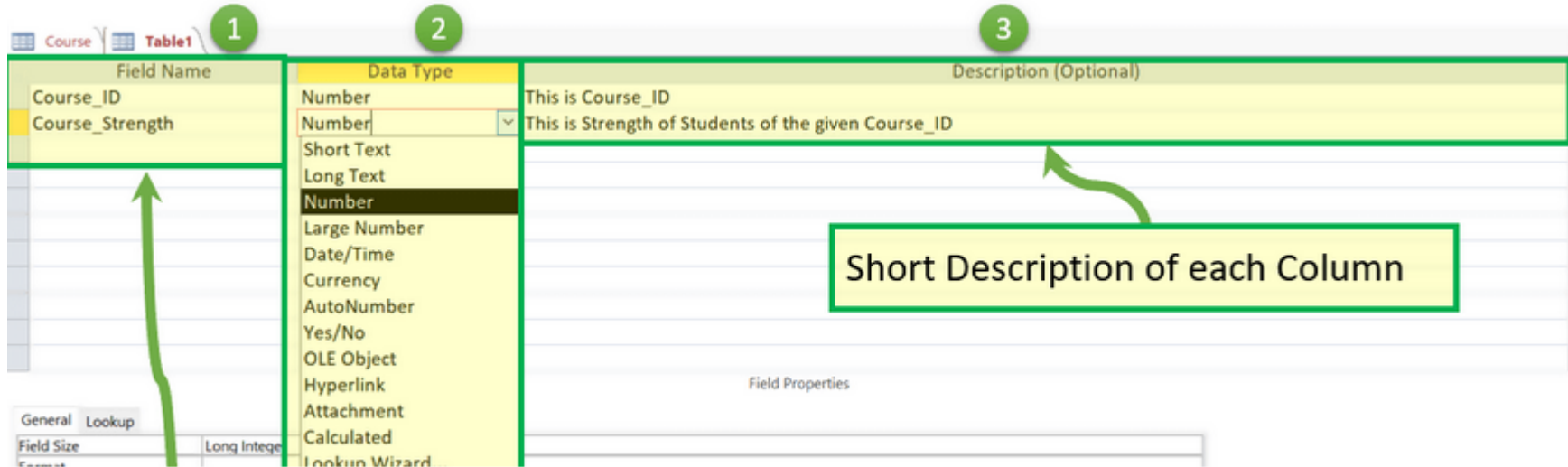
Create Table – Design View

- **Step 1)** First Click Create tab. Then from Tables group, click Table.



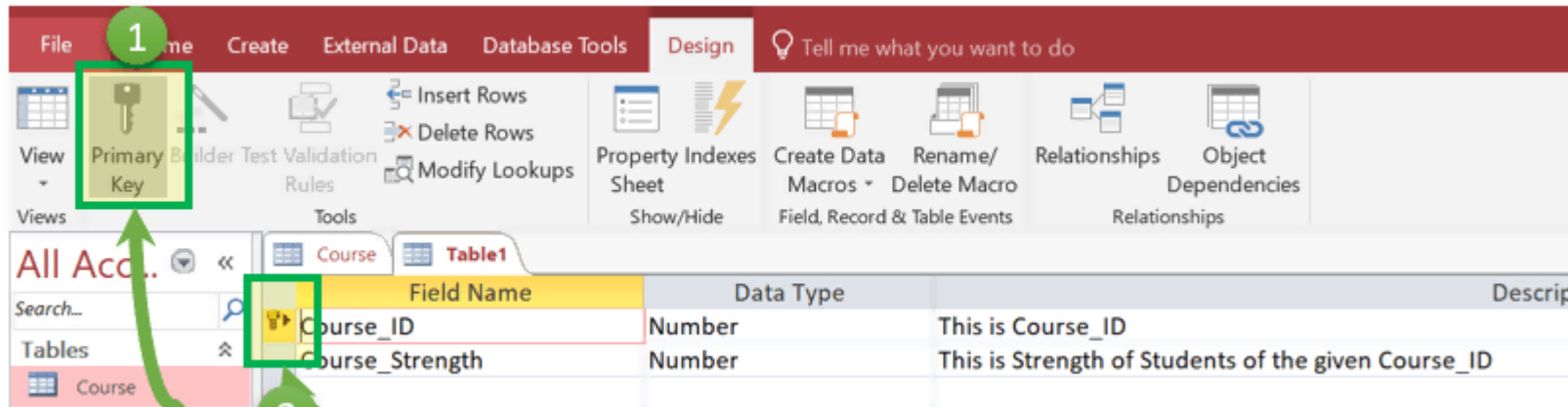
Create Table – Design View

- **Step 2)** Table Dialog box appears. For each Field enter **Filed Name, Data Type** and **Description**.



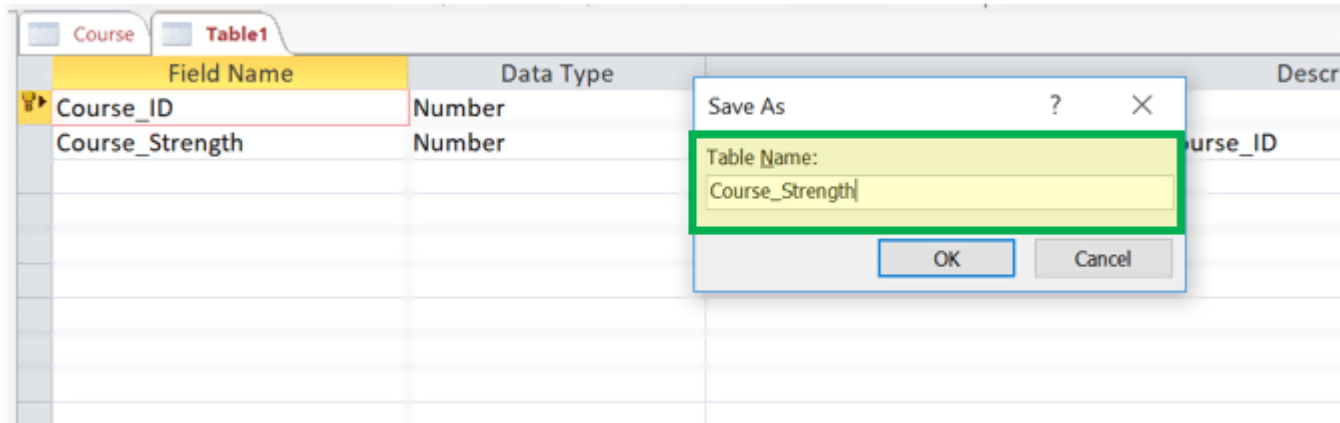
Create Table – Design View

- **Steps 3)** To Add Course_ID as Primary Key, select it and Click on 'Primary Key.' Course_Id will be Preceded by KEY ICON as shown below:



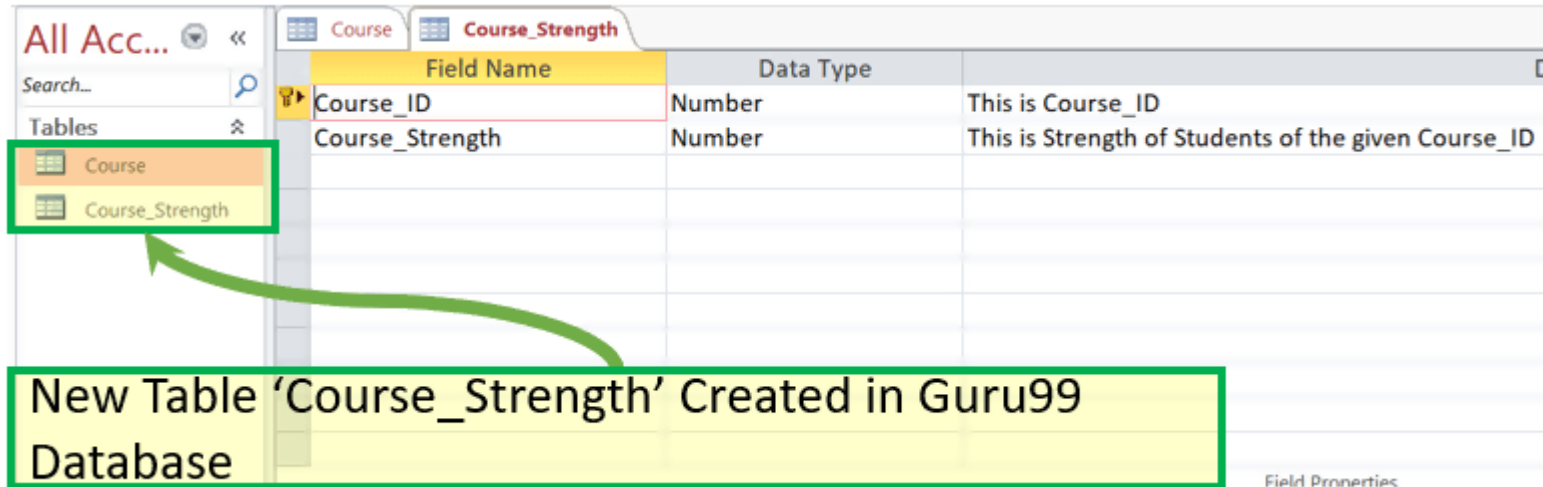
Create Table – Design View

- **Steps 4)** Press 'Ctrl+S.' Enter the Table Name and Click OK



Create Table – Design View

- **Result:**



The screenshot shows the Microsoft Access design view for a database. On the left, the 'Tables' list contains 'Course' and 'Course_Strength', with 'Course_Strength' highlighted by a green box. A green arrow points from a text box below to this table. The main area shows the design view for 'Course_Strength' with two fields: 'Course_ID' (Number) and 'Course_Strength' (Number). The 'Course_Strength' field has a description: 'This is Strength of Students of the given Course_ID'.

Field Name	Data Type	Description
Course_ID	Number	This is Course_ID
Course_Strength	Number	This is Strength of Students of the given Course_ID

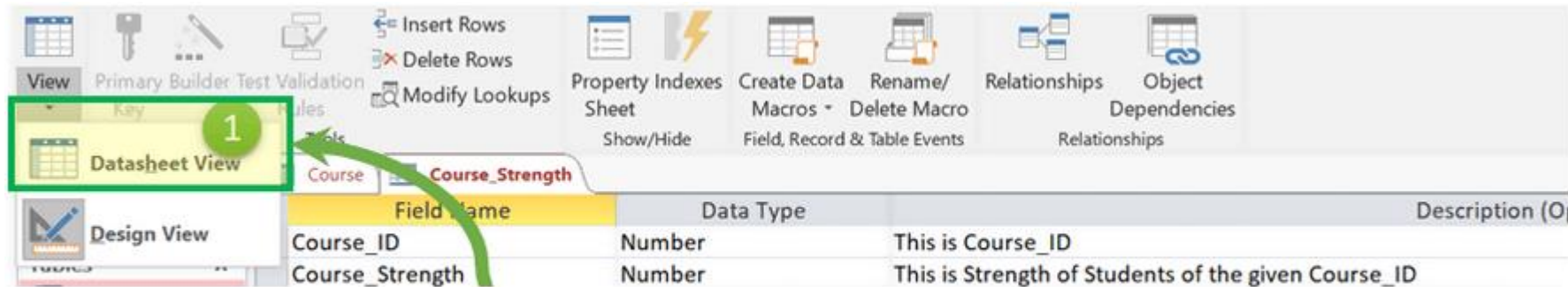
New Table 'Course_Strength' Created in Guru99 Database

How to Switch views between Datasheet and Design

- To switch views, between the datasheet (spreadsheet view) and the design view, click the '**View**' button in the top-left hand corner of the Access Ribbon (shown in the Home/Help/Design Tabs). And Click on the View you need to display.
- For Example: Let's Assume You want to switch to the newly created table 'Course_strength' from 'Design view' to 'Data Sheet' View.

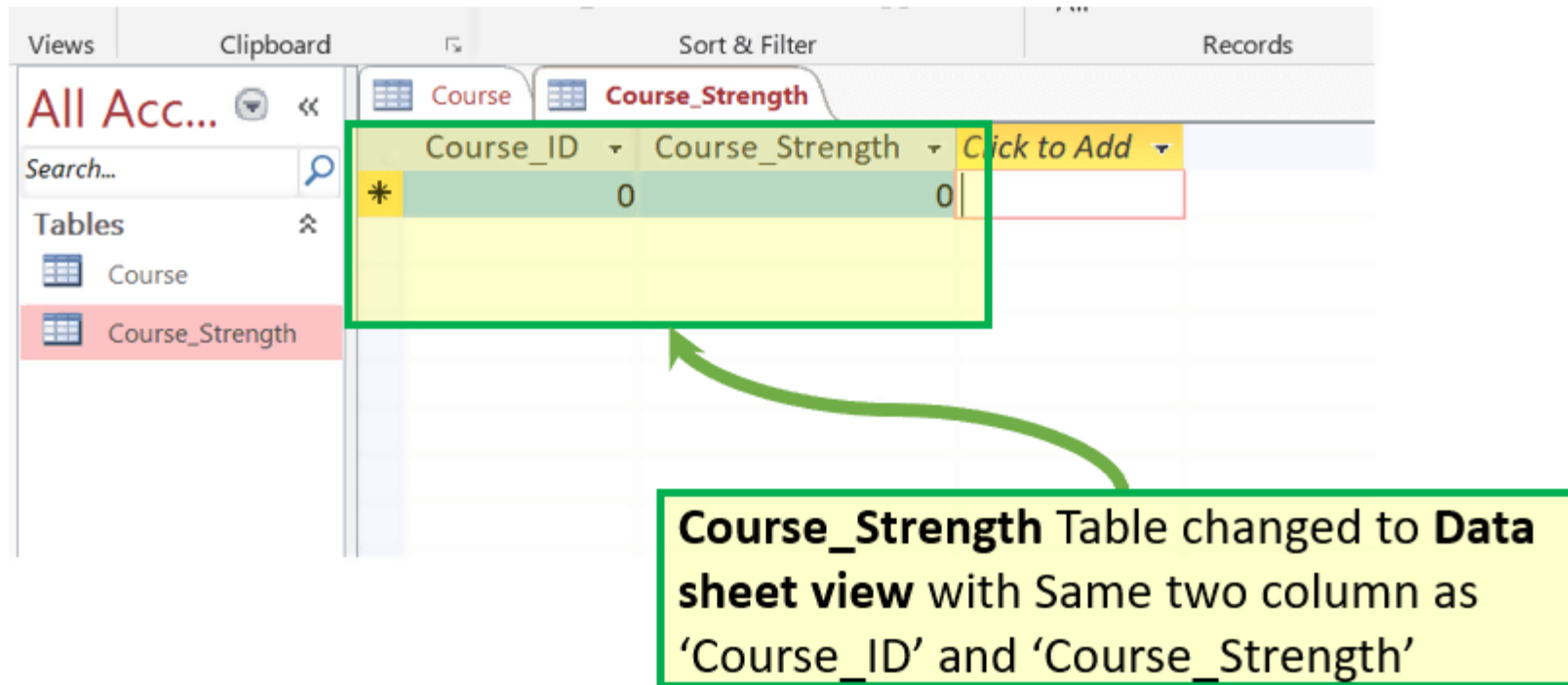
How to Switch views between Datasheet and Design

- **Step 1)** Click on Datasheet View



How to Switch views between Datasheet and Design

- **Result:**



The screenshot shows the Microsoft Access interface. On the left, the 'Tables' list includes 'Course' and 'Course_Strength'. The 'Course_Strength' table is selected and displayed in Datasheet view. The table has two columns: 'Course_ID' and 'Course_Strength'. The first row contains the value '0' in both columns. A green box highlights the first row, and a green arrow points from a text box below to this row.

Course_ID	Course_Strength
0	0

Course_Strength Table changed to **Data sheet view** with Same two column as 'Course_ID' and 'Course_Strength'

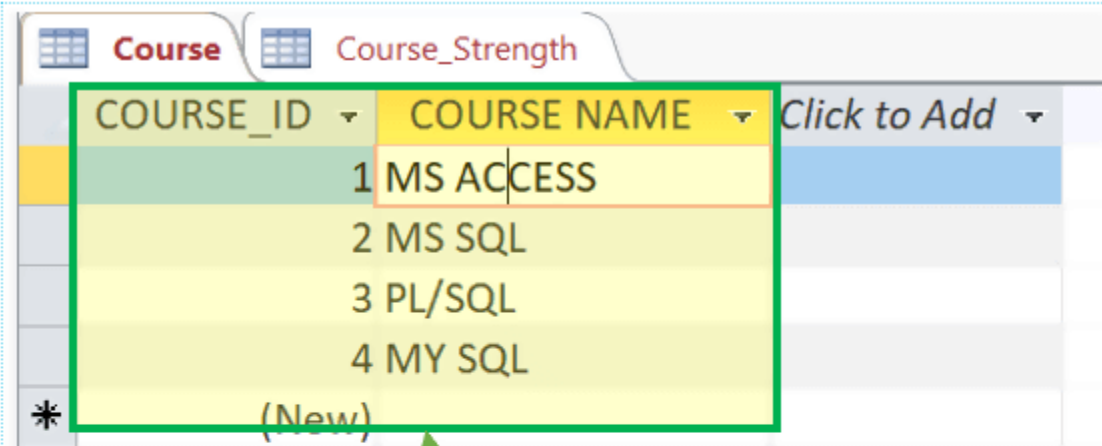
MS Access- Adding Data

- Two ways to add data.
- Datasheet view and other
- From form.
- There is no need to save the data manually. Once you click other rows, access automatically saves the data.
- The only thing you need to take care while adding data is, you should enter the correct data type, i.E., If column is defined as number we can add only number and not text.

MS Access- Adding Data

- Step 1) Select the 'Course' table
- Steps 2) Select the Datasheet view option in the ribbon and add some data by entering the values in It. Updated Data will be Autosaved.

MS Access- Adding Data



The screenshot shows the MS Access interface with the 'Course' table selected. A dropdown menu is open for the 'COURSE NAME' field, displaying a list of course names: '1 MS ACCESS', '2 MS SQL', '3 PL/SQL', and '4 MY SQL'. Below these options is a '(New)' option, which is highlighted by a green box. A green arrow points from a text box below to the '(New)' option.

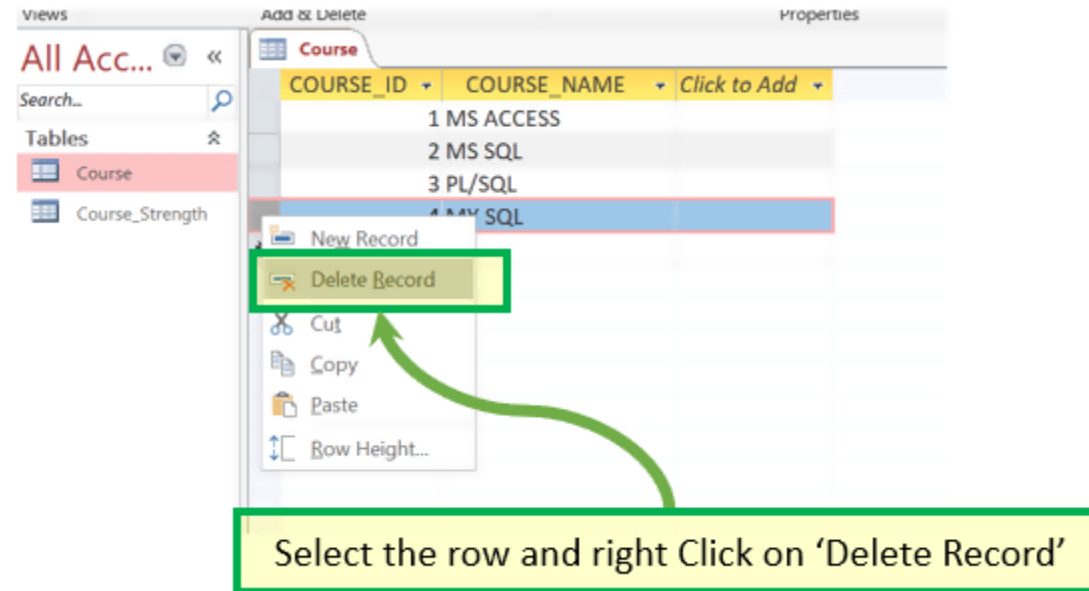
COURSE_ID	COURSE NAME	Click to Add
1	MS ACCESS	
2	MS SQL	
3	PL/SQL	
4	MY SQL	
*	(New)	

New **Course name** added by clicking on each row.
Since **Course_ID** is of 'AutoNumber' datatype, it will automatically increment its value by 1

MS Access- Deleting Data

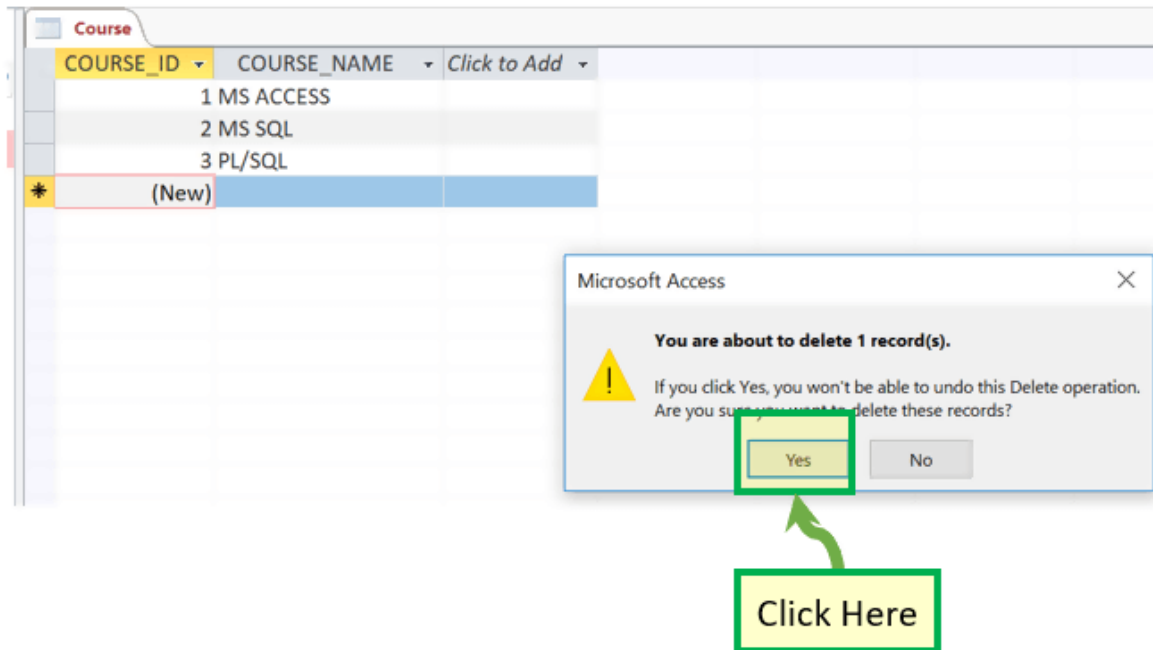
- Now let's say You want to delete 4th row with 'My SQL' as Course name.
- **Step 3)** Select the row by clicking on the leftmost column and Right Click on the row. The options menu will appear with the 'Delete Record' Option

MS Access- Deleting Data



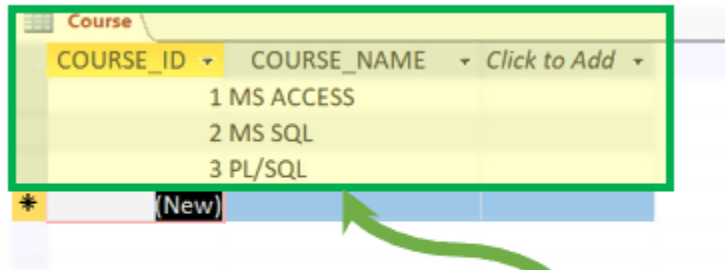
MS Access- Deleting Data

- **Step 4)** Popup Window will appear to confirm the deletion of the record. Click '**OK**'



MS Access- Deleting Data

- **Result:**



A screenshot of the MS Access 'Course' table. The table has two columns: 'COURSE_ID' and 'COURSE_NAME'. There is a 'Click to Add' button next to the 'COURSE_NAME' header. The table contains three rows of data: '1 MS ACCESS', '2 MS SQL', and '3 PL/SQL'. Below the table, there is a new row being added, indicated by a green asterisk and the word '(New)' in a blue box. A green arrow points from a text box below to this new row.

COURSE_ID	COURSE_NAME	Click to Add
1	MS ACCESS	
2	MS SQL	
3	PL/SQL	
*	(New)	

4th Row deleted with 'My SQL' as Course Name

Why Use Forms?

- Forms aid data entry and help users understand data
- A form provides a window into the data
- Users can view several records at a time or work with one record at a time
- Data entry in a well-designed form is much easier than data entry into a datasheet

Controls and Properties

- Forms are made up of controls
- A control is a graphic object that displays in a form or report
 - May be bound or unbound
- A bound control displays data from an underlying table or query
- An unbound control is not tied to fields in an underlying table or query
 - Calculated controls are examples of unbound controls
- Each control on a form possesses properties which affect its behavior and appearance
- Form sections and the form itself also possess properties
- Properties can be viewed in the Property Sheet

Designing a Form

- Begin by listing the data you want to enter (or view)
- Decide on the controls you need in order to capture and work with the data
- Select a data source for the form – that is, a table, or a query

Form Types

Detail form	The most basic type. Shows part or all of a single records from a single data source
Datasheet form	Looks like a datasheet, but allows you to remove fields without affecting the data
Split form	Combination detail form and datasheet form. Allows you to view a large amount of data, but work with one record at a time.
Form with Subform	Shows both sides of a one-to-many relationship, allowing you to work with related data.
Multiple Items form	Shows several records at a time from a single data source.

Form Types

Navigation form	Displays a set of tabs each of which contains a separate form or report
Blank form	Opens an empty canvas in Layout view and allows you to select a data source and drag fields onto the form.
Form Design	Same as blank form, except the empty canvas opens in Design view
Form Wizard	A wizard that steps you through the process of creating a form.
More Options choices	Offers options for creating a modal dialog box which is a special form that displays in front of other forms and stays open until a user closes it

Forms with Subforms

- A subform is a form that is inserted into another form
- The primary form is called the main form, and the form that it enclosed in the main form is the subform
- Subforms are useful for showing data from the “many” side of a one-to-many relationship

Form Views

- There are three views for working with forms:
 - Form view
 - Layout view
 - Design view

Form View

- Allows you to read, enter and edit records
- Use this view when you are working with the data
- You cannot modify a form's design in form view



The screenshot shows a web-based form titled "NYCustomers" with a blue header bar. Below the header, there are four input fields arranged vertically. The first field is labeled "FirstName" and contains the text "Helen". The second field is labeled "LastName" and contains the text "Anderson". The third field is labeled "State" and contains the text "NY". The fourth field is labeled "ZIP" and contains the text "11552". The form is displayed within a window that has a tab labeled "NYCustomers" at the top.

FirstName	Helen
LastName	Anderson
State	NY
ZIP	11552

Layout View

- Show guides that align controls
- Each control displays real data
- Can use tabular or stacked layout
- Allows you to adjust the form's design while looking at real data

CustomerID	FirstName	LastName	State	ZIP
1012	Helen	Anderson	NY	11552
1035	Anna	Billings	NY	11793
1042	Sarah	Armstrong	NY	11558

Design View

- Provides a detailed view of the structure of the form
- Does not show any data
- Allows you to add a wider variety of controls, than does Layout view
- Allows you to resize form sections
- Allows you to adjust the size of individual controls

The screenshot displays a form in Design View, organized into three main sections: Form Header, Detail, and Form Footer. The Form Header section contains a single text box labeled "NY Customers". The Detail section is a large area containing several text boxes for data entry: "CustomerID", "FirstName", "LastName", "State", and "ZIP". The Form Footer section is currently empty. The form is overlaid on a grid background, and the text boxes are labeled with their respective field names.

Automated Form Tool – Detail Form

- Select a table or query in the Navigation Pane
- Display the Create tab of the Ribbon
- Click the Form command button

Automated Form Tool – Datasheet Form

- Select a table or query in the Navigation Pane
- Display the Create tab of the Ribbon
- Click the More Forms in the Forms group
- Click Datasheet

Automated Form Tool – Split Form

- Select a table or query in the Navigation Pane
- Display the Create tab of the Ribbon
- Click the Form command button
- Click More Forms in the Forms group
- Click Split Form

Automated Form Tool – Multiple Items Form

- Select a table or query in the Navigation Pane
- Display the Create tab of the Ribbon
- Click the Form command button
- Click More Forms in the Forms group
- Click Multiple Items

Using the Form Wizard

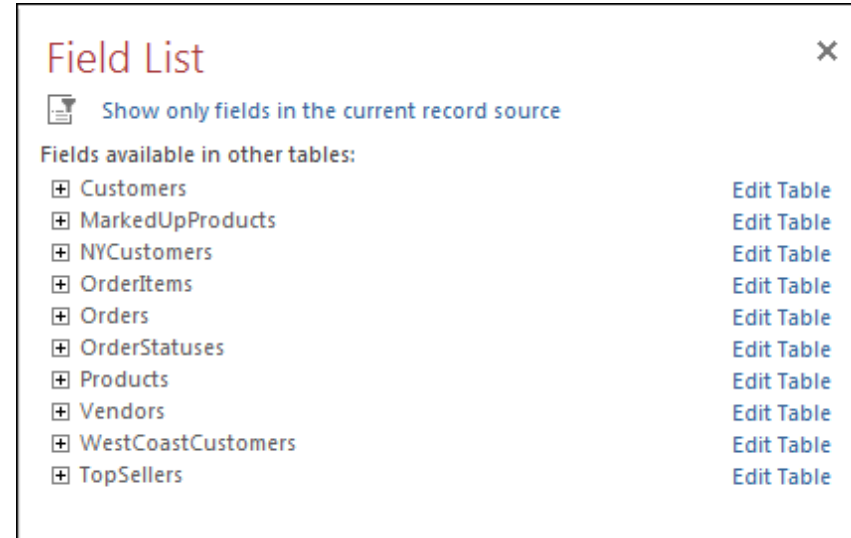
- Using the Form Wizard provides more fine-tuning power when creating forms:
 - You can specify which fields to include on the form
 - You can define how data is grouped and sorted
 - You can use fields from more than one table

Creating Forms from Scratch

- Use a blank form to create a form from scratch
- A blank form is an empty canvas that is not tied to an underlying table or query
- To create a blank form in Layout view, click the Blank Form command button in the Forms group on the Ribbon
- To create a blank form in Design view, click the Form Design command button in the Forms group on the Ribbon
- Drag fields from the Field List pane into the form

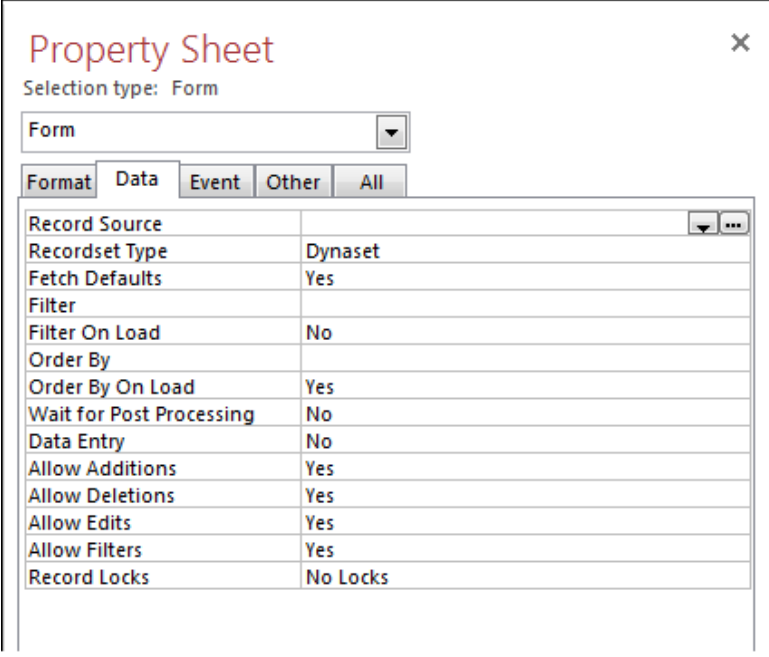
The Field List Pane

- The Field List pane displays the tables in the database
- You can expand any table in the Field List pane and then drag fields into a form



The Property Sheet

- The Property Sheet includes the following tabs:
 - Format
 - Data
 - Event
 - Other
 - All
- To display the Property Sheet, click the Property Sheet button or press F4



The screenshot shows the 'Property Sheet' dialog box with the 'Format' tab selected. The 'Selection type' is set to 'Form'. The dialog contains a table with various properties and their values.

Property Sheet	
Selection type: Form	
Form	
Format Data Event Other All	
Record Source	
Recordset Type	Dynaset
Fetch Defaults	Yes
Filter	
Filter On Load	No
Order By	
Order By On Load	Yes
Wait for Post Processing	No
Data Entry	No
Allow Additions	Yes
Allow Deletions	Yes
Allow Edits	Yes
Allow Filters	Yes
Record Locks	No Locks

The Record Source Property

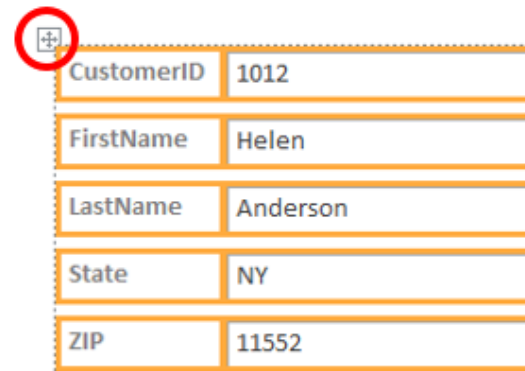
- The Record Source property controls which fields are available for a form because it ties a table, a query or a SQL statement to the form
 - This flexibility allows you to specify multiple data sources for the fields on the form
- The Record Source property applies to the form itself, and is visible in the Property Sheet only when the entire form is selected

The Control Source Property

- The Record Source property controls the data for an individual control
- If you select any control inside a form, the Control Source property appears in the top row of the Data tab in the Property Sheet

Working with a Form in Layout View

- In Layout view controls and their labels are placed in a grid of columns and rows (a layout)
- More than one layout can appear on a form, and each contains a layout selector



CustomerID	1012
FirstName	Helen
LastName	Anderson
State	NY
ZIP	11552

Working with a Form in Layout View

- In Layout view, you can:
 - Select and format controls
 - Move controls
 - Use table functions

Working with a Form in Design View

- In contrast to using layouts, Design view uses a design grid along which you can align individual controls
- Design view allows you to format each control individually, giving you granular control over size and positioning
- If you open a form that has been created by one of the automated form tools, or by the form wizard, you first have to remove controls from the layout if you want to work with them independently

Creating Forms Using Application Parts

- A variety of blank forms display at the top of the Application Parts gallery
- The forms are not tied to database data, but include placeholders for fields

Deleting Forms

- Forms can be deleted in the Navigation Pane
- Keep in mind that other database objects may depend on a form

Adding Controls from the Ribbon

- You can add unbound controls to a form from the Ribbon
- Added controls can be bound to a field by setting the control's Control Source property
- The controls that initially appear in the Controls group vary depending on the view you are using



Adding Controls in Layout View

Select	Clicking this button allows you to select one or more controls on the form.
Text Box	Inserts a text box control which allows the user to enter text, numeric, or date values into a field.
Label	Inserts a label control which is an unbound control typically used to display titles and other descriptive text on the form.
Button	Inserts a command button and launches the Command Button Wizard, which steps you through the process of formatting the command button and creating the code that will make the button function.
Tab Control	Inserts a tab control which allows you to group controls on separate pages in a form. This is useful when you have more controls than you have space available on the form.

Adding Controls in Layout View

Hyperlink	Inserts a hyperlink control. The hyperlink control can be clicked and followed when a user views the form in Form view.
Web Browser Control	Inserts a control that displays Web browser content within a control on the form.
Navigation Control	Inserts a navigation control that allows you to switch easily between the forms and reports in a database. A navigation control is not the same as a tab control: the former will load a different form or report for each tab selected; the latter displays a different set of controls within the same form.
Combo Box	Inserts a combo box (combination text box/drop-down list) into a form. You can type values into the text box or select a value from the drop-down list.
List Box	Inserts a list box into a form that displays values from which a user can choose. The list is always open.

Adding Controls in Layout View

Check Box	Inserts a check box control into a form. A check box control includes a label and a yes/no field.
Attachment	Inserts an attachment field.
Subform/ Subreport	Embeds a subform into a form (or a subreport into a report). The subform control can be thought of as a “view” of another table or query. The subform control provides properties that link the data displayed in the control to the data on the main form.
Image	Inserts an image into a form. When you insert an image using an image control, the image does not change as you move from one record to another. An image control is frequently used to display a corporate or product logo.

Adding Controls in Design View

Option Group	Adds a control that contains a set of related buttons that work together to provide a limited set of choices for the user. The entire option group maps to one field in a database record.
Insert Page Break	Inserts a page break into a form for when it is printed.
Chart	Inserts a chart into a form or report and starts the Chart Wizard, which steps you through the process of creating a chart based on a table or query.
Line	Inserts a line as a graphic element into a form or report. It is used primarily for aesthetic value, such as visually separating one area of a form or report from the rest.
Toggle Button	A graphic element that appears either pushed in or not pushed in. Like the check box, it is used for binary values: yes/no, on/off, 0/1. The Property Sheet button in the Ribbon is an example of a toggle button.

Adding Controls in Design View

Rectangle	Inserts a rectangle as a graphic element into a form. This is also used primarily for aesthetic value to visually separate one area of a form or report from the rest.
Unbound Object Frame	Inserts an object such as a spreadsheet, chart, slide or document into a form. The object is not tied (bound) to any particular record in the database.
Option Button	A graphic element previously called a radio button. Like the check box and toggle button, it is used for binary values: yes/no, on/off, 0/1.
Bound Object Frame	Inserts a frame that will contain a bound object, such as a picture, associated with a particular record in a database.

Aligning and Sizing Controls

- You can align controls with the Horizontal and Vertical rulers
- You can align controls to the design grid
- You can align controls relative to one another
- You can also size controls relative to one another

Adding a Subform Control

- You can create a form and subform simultaneously using the Form Wizard
- You can also use the Subform/Subreport tool to create a subform control

The Visible Property

- The Visible property determines whether a control is visible in a form
 - At times, hiding controls can be advantageous
- The Visible property has two possible settings – Yes and No

Manually Creating and Linking a Subform

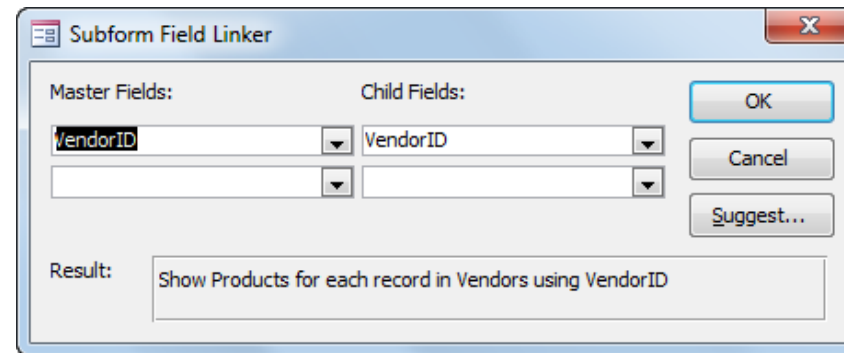
- Access will automatically link a subform to a main form under the following conditions:
 - The main form is based on a table that serves as the “one” side of a one-to-many relationship, and
 - The table on which the main form is based serves as the “one” side of only one, one-to-many relationship

Subform Terminology

Subform control	The control that embeds a subform into a main form. The subform control provides properties which allow you to link the data displayed in the subform to the data on the main form.
Source Object property	The property of the subform control that provides the name of the subform itself. The subform control is a placeholder on the main form and the named subform displays within the subform control.
Link Child Fields property	The property of the subform control that specifies which field or fields in the subform link the subform to the main form.
Link Master Fields property	The property of the subform control that specifies which field or fields on the main form link the main form to the subform.

To manually insert and link a subform:

- Open the main form in either Layout view or Design view
- Drag the subform from the Navigation Pane into the main form
- Click the Link Master Fields or Link Child Fields row in the Property Sheet
- Click the Build button to open the Subform Field Linker dialog box and specify the appropriate linking fields



Using Lookup Fields on Forms

- Forms can be made user-friendly by adding lookup fields where appropriate
- Drag the Combo Box control onto a form
 - If the Control Wizards are activated, the Combo Box wizard will guide you through the creation of the combo box

Line Item	Item ID	Description	Quantity	Unit Price	Extended Price
1	40	Fieldstone stepper garden path	1	175.50	\$175.50
2	5	Garden Hose Vinyl 50 FT	2	15.25	\$30.50
3	17	Rosemary seed, organic	2	3.68	\$7.36
4	22	Hyssop seed, organic	3	3.41	\$10.23
5	16	English lavender seed, organic	3	3.39	\$10.17
6					
*					

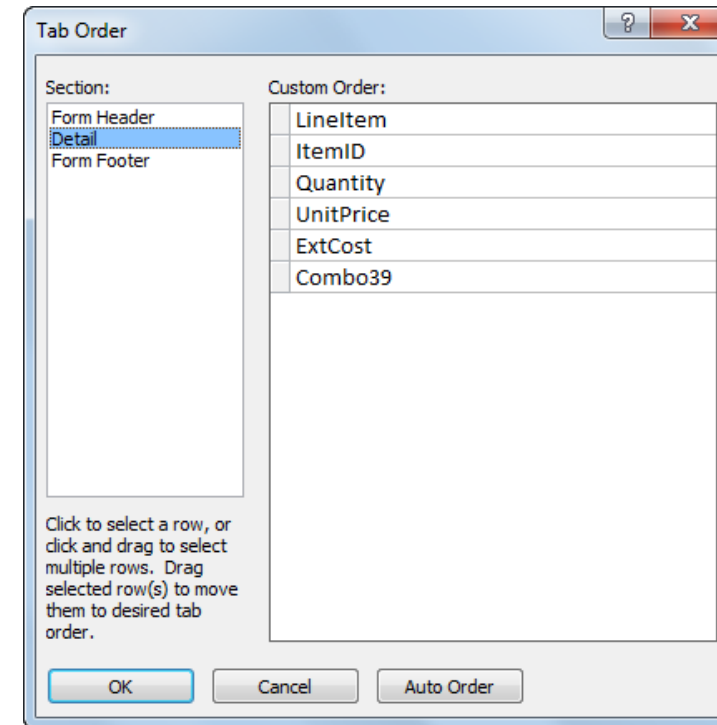
Record: 6 of 6

Angelica seed, organic
Artichoke seed, organic
Basil seed, organic
Catnip seed, organic
Cotton seed, organic
English lavender seed, organic
Fieldstone stepper garden path
Fieldstone, regular grey
Fieldstone, thin grey
Frost pop-up net, 3x3
Frost pop-up net, 3x6
Garden Hose Rubber 100 FT
Garden Hose Rubber 25 FT
Garden Hose Rubber 50 FT
Garden Hose Vinyl 100 FT
Garden Hose Vinyl 25 FT

Order Total \$233.76

Tab Order

- The tab order controls the order in which you can access particular controls by pressing the Tab key
- Use the Tab Order dialog box to view and edit the tab order
- To adjust the order, click a row and drag it to a new position in the dialog box
- Click Auto Order to have Access create a top-to-bottom and left-to-right tab order automatically



Validating Data Entry in Forms

- Techniques for keeping data valid include:
 - Creating validation rules
 - Changing the Tab Stop property
 - Changing the Locked and Enabled properties
 - Specifying default values
 - Setting form properties

Validation Rules

- A validation rule is a criterion that must be met before a record can be saved
- Enter the criterion in the Validation Rule row in the Data tab of the Property Sheet
- You can specify optional validation text that will display as an error message if the validation rule is not met

The Tab Stop Property

- There are two possible settings for the Tab Stop property – Yes and No
- Setting the Tab Stop property to No for a given field causes the cursor to skip the field when the user presses the Tab key
 - Although the user can still click the field, setting the Tab Stop property to No provides a visual clue that the user should not change the data in the field

The Locked and Enabled Properties

- You can use the Locked and Enabled properties to restrict user access to particular controls (and their underlying table or query fields)
- The Enabled property specifies whether a control can receive the focus in Form view
- The Locked property determines whether the data in a control can be edited

Specifying Default Values

- You can specify a default value for a control, and when a new record is added, that field will already contain the value you have specified as the default
- Default values speed data entry and are helpful to inexperienced users who may not know what value to enter
- The Default Value property can be accessed on the Data tab of the Property Sheet

Setting Form Properties

- Form properties control data entry. These properties are:
 - Allow Additions
 - Allow Deletions
 - Allow Edits
 - Data Entry
- Each of these properties can be set to either Yes or No
- Each of these properties is accessed on the Data tab of the Property Sheet for the form

Applying Formatting and Design Options

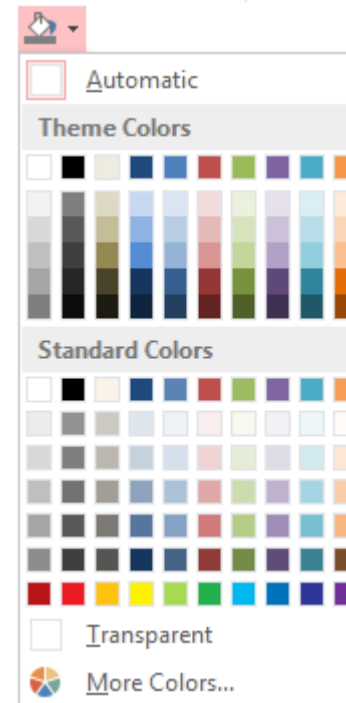
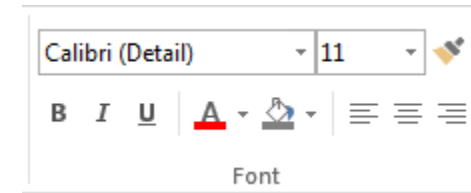
- Design and formatting options for forms include:
 - Working with headers and footers
 - Working with fonts, backgrounds and images
 - Working with themes
 - Formatting forms for printing

Working with Headers and Footers

- The Form Header section appears above the Detail section
- The Form Footer section appears below the Detail section
- Click an option in the Header/Footer group in the Design tab of the Form Design Tools tab. Options include:
 - Logo
 - Title
 - Date and Time

Working with Fonts, Backgrounds and Images

- Use the tools in the Font group to format a form
- A background color can be specified for individual controls or for entire form sections



Working with Fonts, Backgrounds and Images

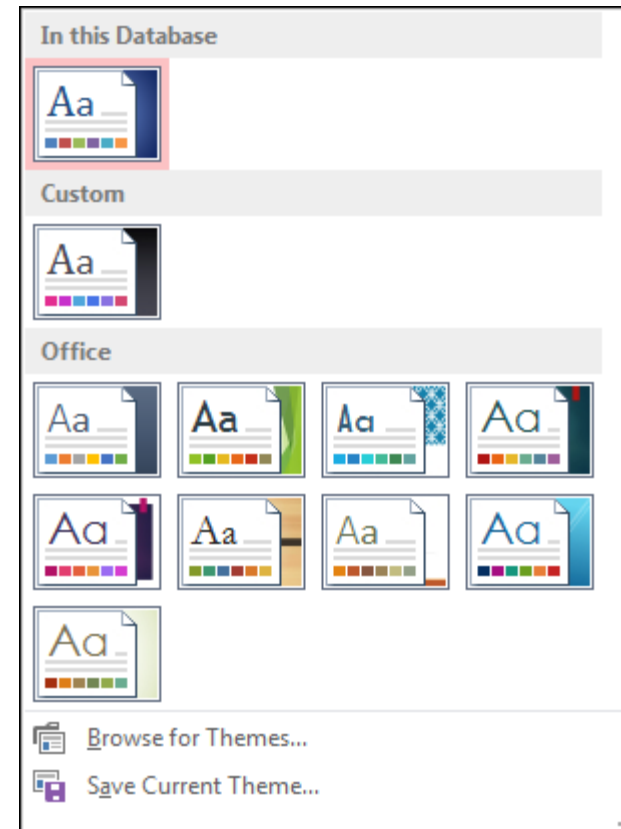
- Background images display behind the form's controls
- To insert a background image
 - Click the Background Image button on the Ribbon
 - Select an image from the Image Gallery, or
 - Click the Browse button to open the Insert Picture dialog box
- The appearance and behavior of a background image are controlled by the Picture Type, Picture, Picture Tiling, Picture Alignment and Picture Size Mode properties for the form
 - These properties are accessible on the Format tab of the Property Sheet

Working with Fonts, Backgrounds and Images

- You can also insert an image into a form using the Insert Image button in the Controls group in the Design tab of the Form Design Tools ribbon
- You can also add an Image control to the form
 - An image control is a frame that will display a picture in a form
 - A bound image control displays a picture stored with each record
 - An unbound image control displays the same picture for all records
- An image can be set to embedded, linked or shared in the Property Sheet

Working with Themes

- Themes include font and color scheme settings
- Applying a theme to a database gives all database objects a consistent look
- You can use a built-in theme or save your own specifications as a custom theme
- Only one theme can be applied to a database at any given time

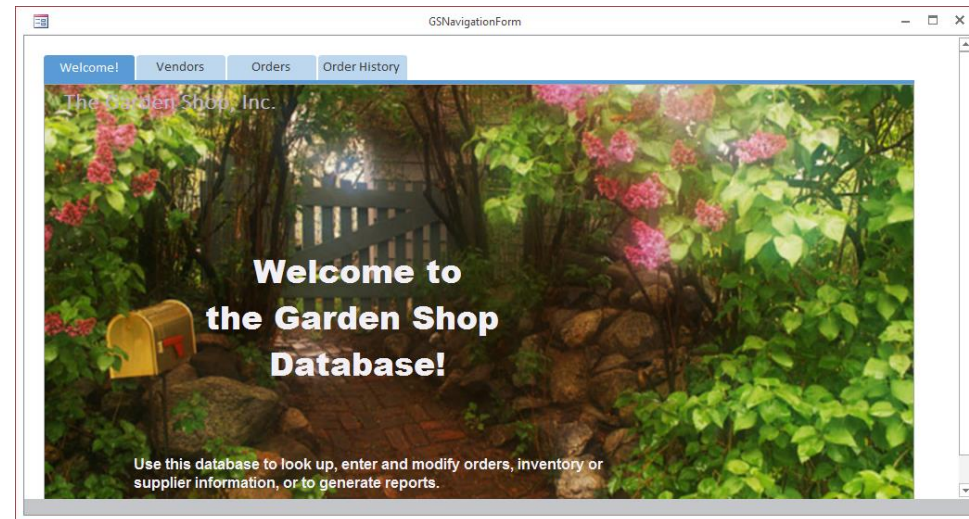


Formatting Forms for Printing

- When you print an Access form, you should keep the following points in mind:
 - You may prefer to sort the records before printing.
 - Access prints all the records for the record source for the form. If you want to print only specific records, select those records and specify to print only the selected records in the Print dialog box, or apply a filter before printing.
 - You will probably need to make a few adjustments to the page orientation and margin settings.
- To print an open form, access the Print tab in Backstage view

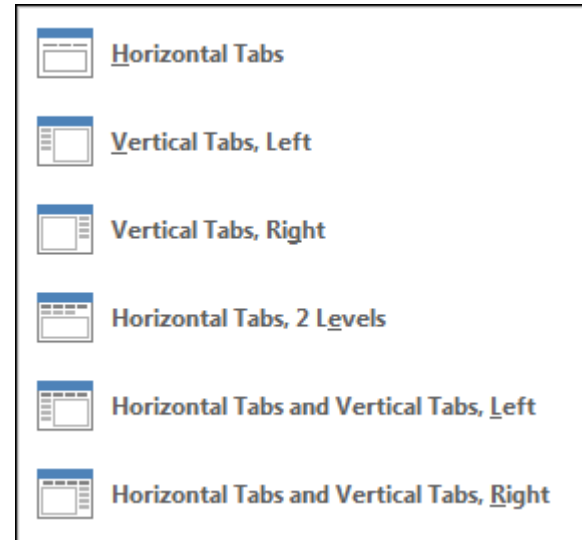
Navigation Forms

- Navigation forms simplify access to data sheets, forms and reports in a database and replace the switchboards of older versions of Access
- Simply click a tab in a navigation form to access the form or report included on the tab



Navigation Forms - Creating

- To create a navigation form
 - Click Navigation on the Ribbon, then click the layout you want to use
 - Drag an existing form or report from the Navigation Pane to the Add New box on the navigation form
 - Continue adding forms and reports as desired
 - Save the navigation form



Closing a Database

- Use the Close command in Backstage view, OR
- Close the application window to close the file and exit Access

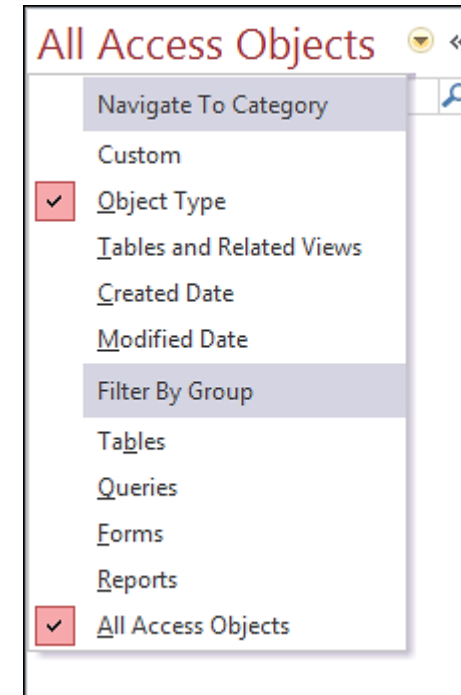
Opening an Existing Database

- Recently used databases appear in the Quick Access list in the Startup screen – simply click the database name to open the file
- You can click the Recent tab to see a list of up to 50 database files you have recently accessed – click a database name to open the file
- You can configure Access to provide quick access links to up to 25 databases at the bottom of the left panel in Backstage view – click a quick access link to open the database
- You can use the Open dialog box to locate and open a database file as well

Using the Navigation Pane

Viewing Objects

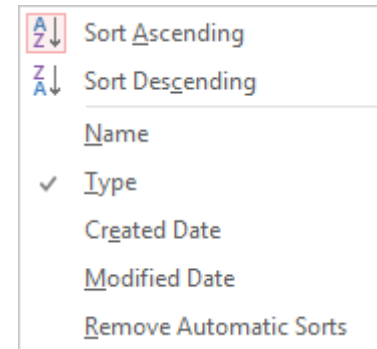
- Select options in the Navigation Pane menu to control which objects display in the Navigation Pane
- By default, objects display by type and all objects appear in the Navigation Pane
- You can filter which objects display
- You can control how objects are grouped or organized



Using the Navigation Pane

Sorting and Filtering Objects

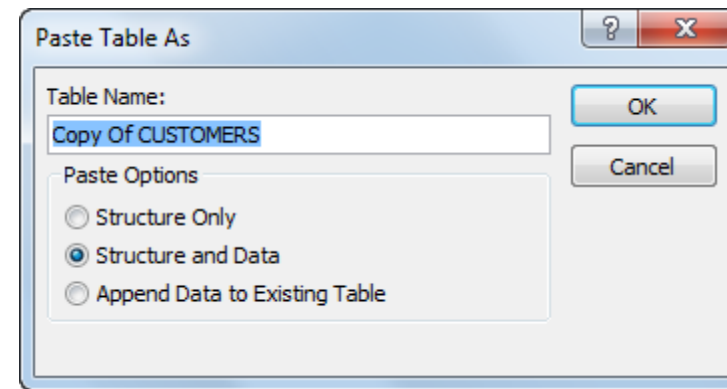
- You can sort and filter objects in the Navigation Pane
- To sort, right-click the object group heading, point to Sort By, then select an option
- To filter objects, select an item in the Filter By Group list



Using the Navigation Pane

Copying Objects

- You can copy and paste objects in the Navigation Pane
- Different objects have different options
- You can also use the Save Object As command to create a copy of the current object with a new name



Using the Navigation Pane

Renaming Objects

- Right-click an object in the Navigation Pane and select Rename in the shortcut menu to open an object name for editing

Using the Navigation Pane

Deleting Objects

- Select an object in the Navigation Pane, then press the Delete key or click the Delete button in the Home tab on the Ribbon
- You can also right-click the object in the Navigation Pane and select Delete in the shortcut menu
- You will be prompted to confirm object deletions
- Take care when deleting objects – make sure that they are not referenced by other objects

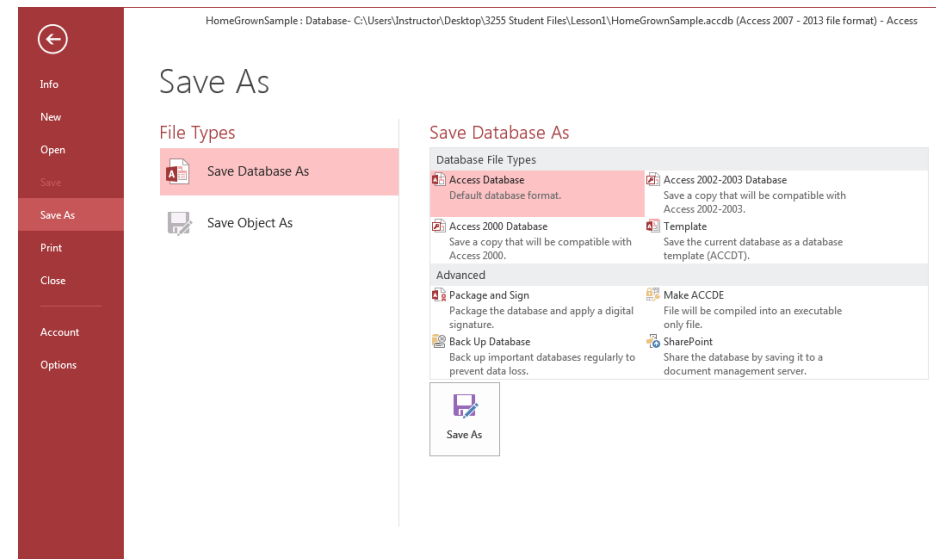
Using the Navigation Pane

Setting Options

- You can create custom categories and groups and set options that control how objects displays in the Navigation Pane
- To create a custom category, right-click the Navigation Pane menu bar, then select **Navigation Options** to open the Navigation Options dialog box
 - Click the **Add Item** button to create a new category
 - Specify a name for the new category, then click **OK**
 - Display the Navigation Pane menu and select the new category, then drag items from the Unassigned Objects list into the Items list so that they will appear in the new category

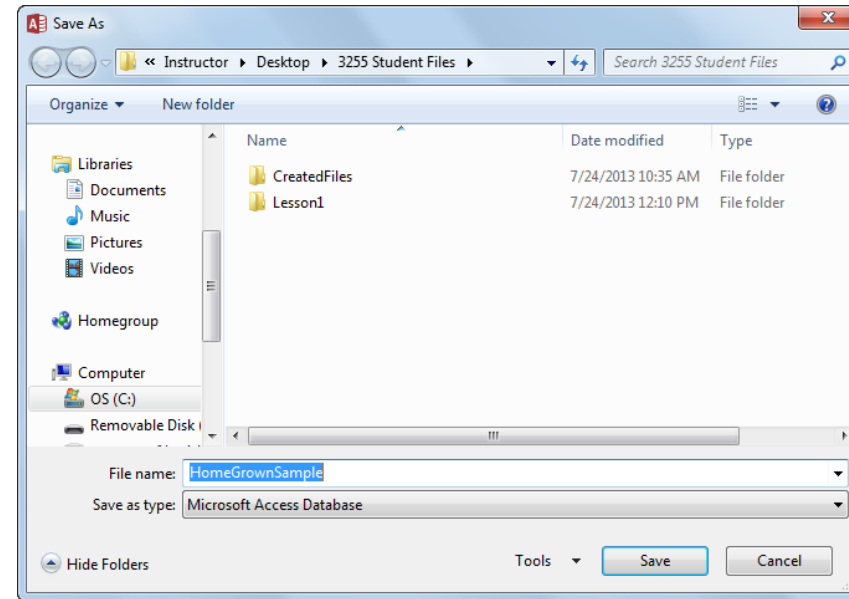
Saving and Exporting Databases

- Use options in the Save As tab in Backstage view to save a database with a different file name or save a database in a different location



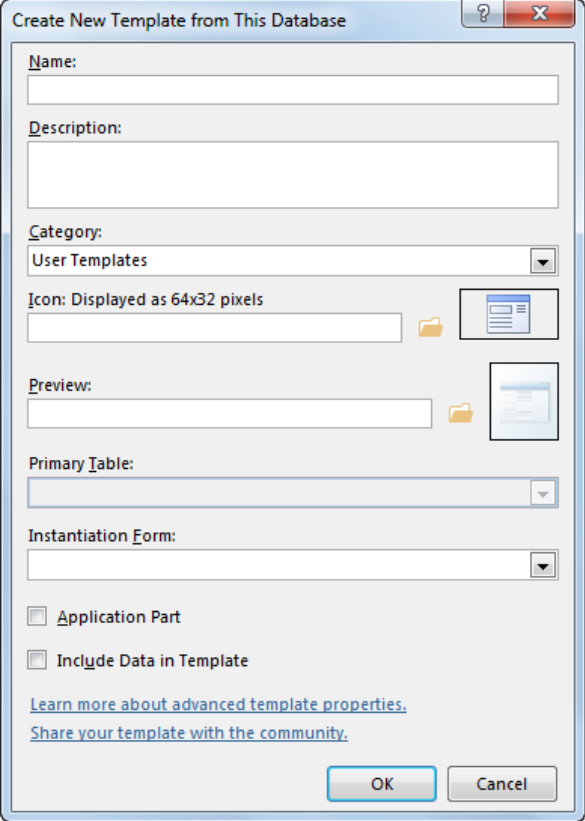
Saving Databases to a Different Location

- Use the Save As dialog box to save a database file to a different location
- Locations can include the local network, an attached storage device, a cloud location or your organization's SharePoint site



Saving a Database as a Template

- On the Save As tab select Template in the Database File Types section, then click Save As to open the Create New Template from This Database dialog box



The dialog box is titled "Create New Template from This Database". It contains the following fields and options:

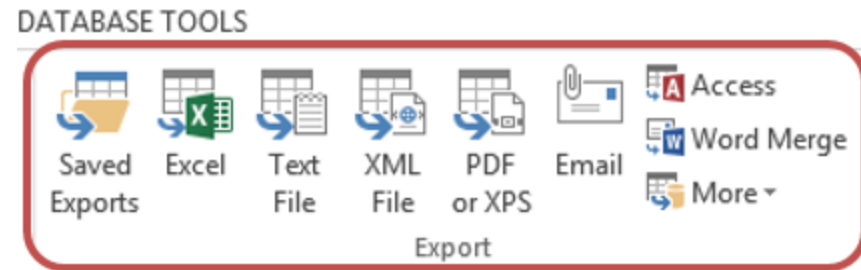
- Name:** A text input field.
- Description:** A text input field.
- Category:** A dropdown menu with "User Templates" selected.
- Icon:** A text input field with the label "Displayed as 64x32 pixels" and a folder icon.
- Preview:** A text input field with a folder icon.
- Primary Table:** A dropdown menu.
- Instantiation Form:** A dropdown menu.
- ☐ Application Part
- ☐ Include Data in Template
- [Learn more about advanced template properties.](#)
- [Share your template with the community.](#)
- OK** and **Cancel** buttons.

Microsoft Access and Backward Compatibility

- Microsoft Access can open .accdb and .mdb database files
- You can convert older files to a newer format using the Save As command
- You can also save an Microsoft Access database to an older format

Exporting Data to Alternate Formats

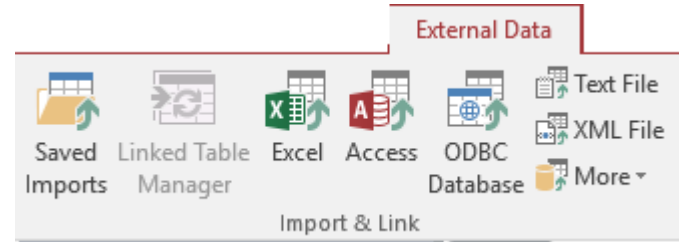
- Excel
- Text file
- XML
- PDF or XPS
- Email
- Access
- Word Merge
- Word (listed under More)



- SharePoint List (listed under More)
- ODBC Database (listed under More)
- HTML Document (listed under More)

Importing Data to Alternate Formats

- Excel
- Text file
- XML
- PDF or XPS
- Email
- Access
- Word Merge
- Word (listed under More)



- SharePoint List (listed under More)
- ODBC Database (listed under More)
- HTML Document (listed under More)

Access Options dialog box contains 11 tabs

General	Contains options for working with Access
Current Database	Contains options that allow you to control how a user can interact with the current database
Datasheet	Contains options that affect the appearance of datasheets
Object Designers	Contains options that control how object designers (the tools that help you create tables, queries, forms, etc.) function
Proofing	Contains settings for AutoCorrect and the spelling checker
Language	Contains settings for specifying default editing, display and Help languages

Access Options dialog box contains 11 tabs

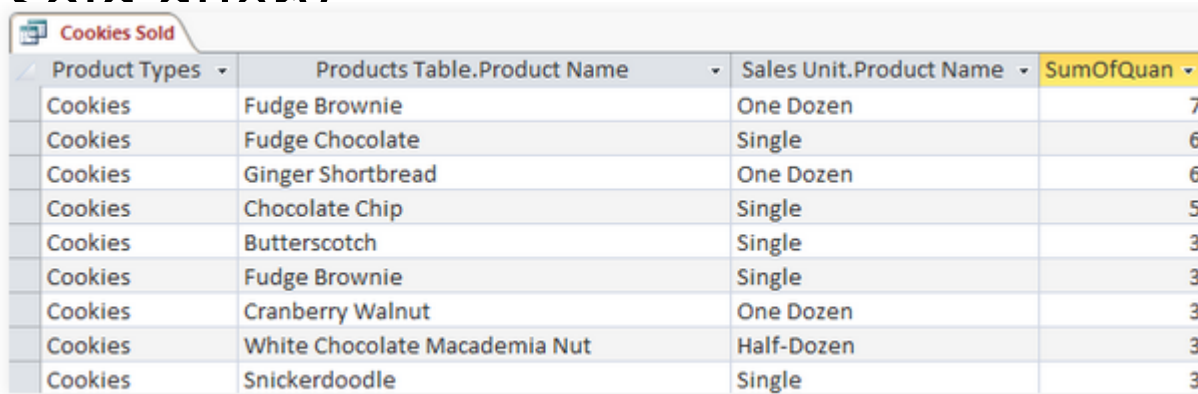
Client Settings	Contains settings that control the behavior of the cursor, default printing margins and default open mode
Customize Ribbon	Contains tools you use to customize the Ribbon
Quick Access Toolbar	Contains tools you can use to customize the Quick Access toolbar
Add-ins	Contains options that allow you to view and control Microsoft Office add-ins
Trust Center	Contains links to privacy statements and a command button that opens the Trust Center, which controls security settings for Access

Microsoft Access - Create a report

- **Reports** give you the ability to present components of your database in an easy-to-read, printable format. Access lets you create reports from both **tables** and **queries**.

Microsoft Access - Create a report

- Step1: Open the table or query you want to use in your report. We want to print a list of cookies we've sold, so we'll open the Cookies Sold query.



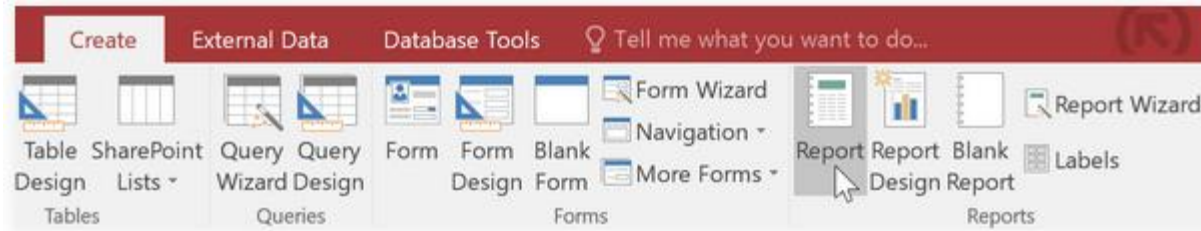
The screenshot shows a table titled 'Cookies Sold' with four columns: 'Product Types', 'Products Table.Product Name', 'Sales Unit.Product Name', and 'SumOfQuan'. The table contains ten rows of data, each representing a different cookie product and its sales unit.

Product Types	Products Table.Product Name	Sales Unit.Product Name	SumOfQuan
Cookies	Fudge Brownie	One Dozen	7
Cookies	Fudge Chocolate	Single	6
Cookies	Ginger Shortbread	One Dozen	6
Cookies	Chocolate Chip	Single	5
Cookies	Butterscotch	Single	3
Cookies	Fudge Brownie	Single	3
Cookies	Cranberry Walnut	One Dozen	3
Cookies	White Chocolate Macademia Nut	Half-Dozen	3
Cookies	Snickerdoodle	Single	3

The Cookies Sold query

Microsoft Access - Create a report

- Step2: Select the Create tab on the Ribbon. Locate the Reports group, then click the Report command.

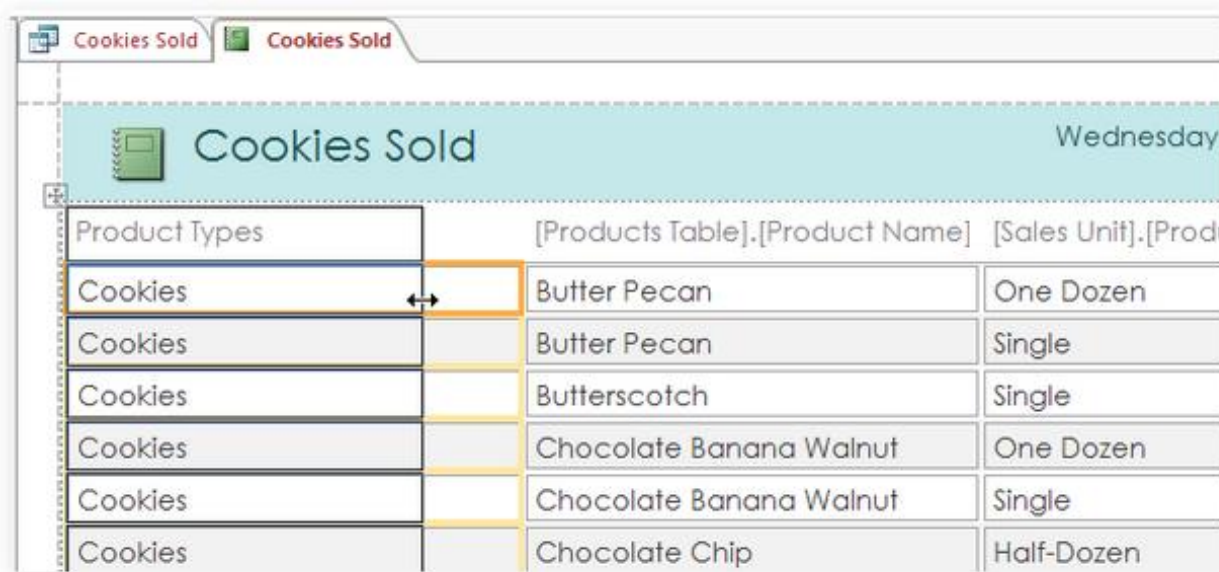


Clicking the Report command

- Step3: Access will create a new report based on your object.

Microsoft Access - Create a report

- Step4: It's likely that some of your data will be located on the other side of the page break. To fix this, resize your fields. Simply select a field, then click and drag its edge until the field is the desired size. Repeat with additional fields until all of your fields fit.



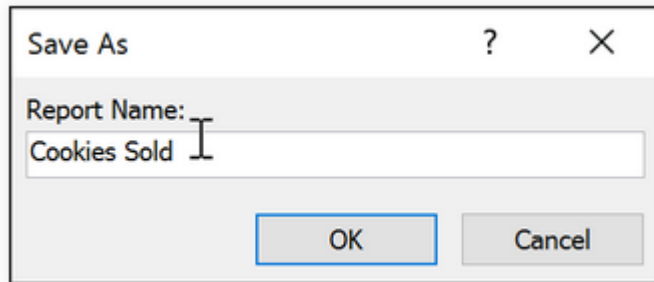
The screenshot shows a Microsoft Access report titled "Cookies Sold". The report has a header section with the title "Cookies Sold" and a date "Wednesday, [Date]". Below the header is a table with three columns: "Product Types", "[Products Table].[Product Name]", and "[Sales Unit].[Product Name]". The table contains six rows of data. The first row is highlighted, and a double-headed arrow is visible on the right edge of the "Product Types" column, indicating it is being resized.

Product Types	[Products Table].[Product Name]	[Sales Unit].[Product Name]
Cookies	Butter Pecan	One Dozen
Cookies	Butter Pecan	Single
Cookies	Butterscotch	Single
Cookies	Chocolate Banana Walnut	One Dozen
Cookies	Chocolate Banana Walnut	Single
Cookies	Chocolate Chip	Half-Dozen

Resizing fields in the report

Microsoft Access - Create a report

- Step5: To save your report, click the Save command on the Quick Access Toolbar. When prompted, type a name for your report, then click OK.

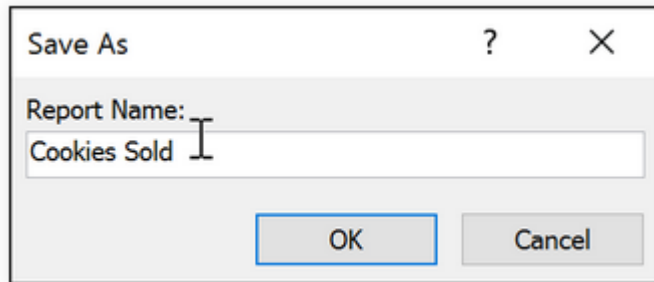


Saving and naming the report

- Just like tables and queries, reports can be sorted and filtered. Simply right-click the field you want to sort or filter, then select the desired option from the menu.

Microsoft Access - Using Queries

- Step5: To save your report, click the Save command on the Quick Access Toolbar. When prompted, type a name for your report, then click OK.



Saving and naming the report

- Just like tables and queries, reports can be sorted and filtered. Simply right-click the field you want to sort or filter, then select the desired option from the menu.

What is a Query?

- Queries are “objects” in Microsoft Access databases
- They provide flexible ways to
 - VIEW data stored in tables
 - manipulate data to make new tables
 - append data from one table onto another

In a Relational Database

- ORGANIZE and STORE Data in TABLES
- ENTER Data in A Data Entry FORM
- VIEW and ANALYZE Data in QUERIES

Queries are Flexible and Powerful

- Create them for a quick look
- Save ones to use regularly
- Use them to edit or clean data
- Use them to assign codes to text data
- Use them to manage complex forms
- Use them to manage reports

Basic Types of Queries

- SELECT queries (the most common type)
- MAKETABLE queries
- APPEND queries
- (other kinds we will not use)
- Today we focus on SELECT queries

What a SELECT Query Does

- Lets you SELECT data to view
- You can pick a table to view
- Then choose what FIELDS to view
- Sort records by one or more fields
- Limit which records you see
- Combine data from multiple tables

To Make a One Table Query

- Select the Query Object Menu
- Choose New to Design a new Query
- Choose the table with the data
- Drag fields from the table to display
- Toggle to Datasheet view to see results

Ways to Manipulate Queries

- Choose to sort one or more fields (from left)
- Use criteria to limit what data displays
- Add more fields from the table
- Add another table to see combined data

Know What You Are Doing

- A simple select query just DISPLAYS
- The data remains in its table, unchanged
- But in a query in datasheet view
 - you CAN make a change to the data in a field
 - that change is written to the table
- NB:
 - make changes when you intend to
 - but don't do it accidentally. THINK first!

To Save or Not to Save

- You can make a quick query and not save
- If you save it, give it a meaningful name
- You can quickly change in design view
 - add or drop fields in the query
 - change the way it sorts
 - change the order of fields (to get the sort right)
 - change criteria for what data displays
- You can save the changes or not save them

Learn More About Queries

- Some Magic Words for Queries
- Use a Query to Code Text
- Put Two Tables into one Query

Some Magic Criteria Words

- For numeric fields
 - use < > to limit the range
 - use = to specify a particular number or date
 - use >0 to drop blanks
- For text fields
 - IS NOT NULL omits any blanks
 - IS NULL shows only the blanks
 - " " selects records with specific text
- These change the VIEW not the records

Use a Query to Help Code Text

- Add a code field to the table if it doesn't have one
- Make a query with the text field and the code field next to each other
- Sort the text field alphabetically
- This should group records with the same text
- Identify groups to assign codes to
- Write the code number in those records
- Check for variations and misspellings and fix
- Add the new codes to lookup table and define them

Concepts for Two Table Queries

- The tables have to have a LINKING FIELD
- The link in the query is called a JOIN
- What data displays is determined by
 - The nature of the relation between the tables (one to one, one to many) AND
 - the type of join
- Think through the logic
 - what do you want to display?
 - did you get what you expected?

What's a JOIN?

- A JOIN is the link between the tables
- tables may have a One-to-Many relation
- The type of join says what records display
- An INNER JOIN only shows records that link directly on both sides
- An OUTER JOIN shows all records on one side, and only the linked ones on the other

Why Do You Need Them?

- Often you just want to see the linked records
 - to see ONE side context for MANY side records
 - to check particular subsets of linked records
 - to look at the names of a lookup code
- Sometimes you need an OUTER JOIN
 - To see what's missing on one side
 - To see WHICH records have the link
 - To clean data or match two sets of data

One-to-Many Queries

- If you join two tables in a query
- That have a one-to-many relation
- The number of RECORDS in the query
- Is the number on the MANY SIDE
- Not the number on the ONE SIDE
- Because the ONE SIDE information
- Is duplicated for each MANY SIDE record

Making Two Table Queries

- Select two tables for the query
- Link the matching ID field (if Access doesn't do it automatically)
- Drag the fields you want to display
- Set the sort and any criteria
- Specify the type of JOIN
- Save, check datasheet, adjust as needed